Advanced Linear Algebra II

Yifan Li

August 2026

Contents

Introduction **7**
0.1 Beyond Diagonalization . . . 7
0.2 From Algebraic Structure to Geometric Structure . . . 7
0.3 Spectral Structure . . . 8
0.4 Bilinear and Quadratic Forms . . . 8
0.5 Matrix Decompositions . . . 9
0.6 Different Ways to Simplify a Matrix . . . 9
0.7 Scope of the Second Volume . . . 10
0.8 Organization of This Volume . . . 10

1 Jordan Normal Form . . . 12
1.1 From Diagonalization to Jordan Form . . . 13
1.2 Polynomial Relations and Annihilating Polynomials . . . 13
1.3 The Minimal Polynomial . . . 14
1.4 Minimal Polynomial and Similarity . . . 15
1.5 A Diagonalization Criterion . . . 15
1.6 Generalized Eigenvectors . . . 16
1.7 Stabilization of Kernels . . . 17
1.8 Primary Decomposition . . . 17
1.9 Nilpotent Operators . . . 18
1.10 Jordan Chains . . . 19
1.11 Jordan Blocks . . . 20
1.12 Jordan Chains and Jordan Blocks . . . 20
1.13 Jordan Normal Form . . . 21
1.14 The Structure of a Jordan Form . . . 21
1.15 Diagonalization Revisited . . . 22
1.16 Determining Jordan Blocks from Kernel Dimensions . . . 22
1.17 Example: Recovering the Jordan Structure . . . 23
1.18 Constructing a Jordan Basis . . . 24
1.19 Characteristic and Minimal Polynomials from Jordan Form . . . 24
1.20 Uniqueness of Jordan Normal Form . . . 25
1.21 Similarity Classification by Jordan Form . . . 25
1.22 Functions and Powers of Jordan Blocks . . . 26
1.23 The Exponential of a Jordan Block . . . 26
1.24 The Structural Meaning of Jordan Form . . . 27

2 Inner Product Spaces and Gram–Schmidt Orthogonalization . . . 28
2.1 Inner Products . . . 29
2.2 Standard Inner Products . . . 29
2.3 Norms . . . 30
2.4 Cauchy–Schwarz Inequality . . . 31
2.5 Triangle Inequality . . . 31
2.6 Angles . . . 32
2.7 Orthogonality . . . 32
2.8 Pythagorean Theorem . . . 33
2.9 Orthonormal Bases . . . 33
2.10 Gram Matrices . . . 34

2.11 Orthogonal Complements . . . 35
2.12 Orthogonal Projection onto a Subspace . . . 35
2.13 Projection Formula . . . 36
2.14 Best Approximation Property . . . 36
2.15 Gram–Schmidt Orthogonalization . . . 37
2.16 Example of Gram–Schmidt . . . 38
2.17 Existence of Orthonormal Bases . . . 39
2.18 Orthogonal and Unitary Matrices . . . 39
2.19 Orthogonal and Unitary Transformations . . . 40
2.20 QR Decomposition . . . 40
2.21 Projection Matrices . . . 41
2.22 Least Squares and Orthogonal Projection . . . 42
2.23 Least Squares via QR Decomposition . . . 43
2.24 The Geometry Introduced by an Inner Product . . . 43

3 Adjoint and Normal Operators . . . 44
3.1 The Adjoint of a Linear Map . . . 45
3.2 Riesz Representation in Finite Dimensions . . . 45
3.3 Basic Properties of the Adjoint . . . 46
3.4 Matrix Representation of the Adjoint . . . 46
3.5 Kernel and Image Relations . . . 47
3.6 Self-Adjoint Operators . . . 48
3.7 Skew-Adjoint Operators . . . 49
3.8 Isometries . . . 49
3.9 Unitary and Orthogonal Operators . . . 50
3.10 Positive Operators . . . 50
3.11 Normal Operators . . . 51
3.12 Norm Equality for Normal Operators . . . 51
3.13 Kernels of a Normal Operator and Its Adjoint . . . 52
3.14 Eigenvectors of Normal Operators . . . 52
3.15 Invariant Subspaces and Orthogonal Complements . . . 53
3.16 Normal Matrices . . . 54
3.17 Unitary Similarity . . . 55
3.18 Schur Triangularization . . . 55
3.19 Normality and Triangular Matrices . . . 56
3.20 Toward the Spectral Theorem . . . 56

4 Spectral Theorems . . . 57
4.1 The Spectral Problem . . . 58
4.2 The Spectral Theorem for Normal Operators . . . 58
4.3 Orthogonal Decomposition into Eigenspaces . . . 59
4.4 Spectral Projections . . . 60
4.5 Polynomial Functions of a Normal Operator . . . 60
4.6 Self-Adjoint Operators . . . 61
4.7 Real Symmetric Matrices . . . 61
4.8 Hermitian Matrices . . . 62
4.9 Skew-Hermitian Matrices . . . 62
4.10 Unitary Operators . . . 62
4.11 Orthogonal Matrices . . . 63

4.12 Positive Semidefinite Operators . . . 63
4.13 Square Roots of Positive Operators . . . 64
4.14 Rayleigh Quotient . . . 64
4.15 Extremal Characterization of Eigenvalues . . . 65
4.16 Restriction to Orthogonal Complements . . . 65
4.17 Courant–Fischer Min–Max Principle . . . 66
4.18 Spectral Norm of a Hermitian Matrix . . . 66
4.19 Spectral Decomposition of a Matrix . . . 67
4.20 Spectral Theorem versus Jordan Theory . . . 67
4.21 Summary of the Spectral Theorem . . . 68

5 Bilinear Forms . . . 69
5.1 Bilinear Forms . . . 70
5.2 Matrix Representation . . . 70
5.3 Gram Matrices of Bilinear Forms . . . 71
5.4 Change of Basis and Congruence . . . 71
5.5 Similarity versus Congruence . . . 72
5.6 Transpose of a Bilinear Form . . . 73
5.7 Symmetric Bilinear Forms . . . 73
5.8 Alternating Bilinear Forms . . . 73
5.9 Symmetric and Alternating Parts . . . 74
5.10 Left and Right Radicals . . . 75
5.11 Degenerate and Nondegenerate Forms . . . 75
5.12 Orthogonality for Bilinear Forms . . . 76
5.13 Nondegenerate Subspaces . . . 76
5.14 Orthogonal Direct Sums . . . 77
5.15 Diagonalization of Symmetric Bilinear Forms . . . 77
5.16 Constructing an Orthogonal Basis . . . 78
5.17 Rank of a Bilinear Form . . . 78
5.18 Classification over Different Fields . . . 79
5.19 Alternating Forms and Their Matrix Structure . . . 79
5.20 Bilinear Forms and Inner Products . . . 80
5.21 From Bilinear Forms to Quadratic Forms . . . 80

6 Quadratic Forms . . . 81
6.1 Quadratic Forms . . . 82
6.2 From Quadratic Forms to Symmetric Bilinear Forms . . . 83
6.3 Matrix Representation . . . 83
6.4 Change of Variables . . . 84
6.5 Diagonalization by Congruence . . . 84
6.6 Completing the Square . . . 84
6.7 Congruence by Elementary Operations . . . 85
6.8 Rank of a Quadratic Form . . . 85
6.9 Real Quadratic Forms . . . 86
6.10 Positive and Negative Indices . . . 86
6.11 Sylvester's Law of Inertia . . . 87
6.12 Signature . . . 87
6.13 Geometric Meaning of Inertia . . . 88
6.14 Definite and Indefinite Quadratic Forms . . . 88

6.15 Eigenvalue Criterion for Definiteness . . . 89
6.16 Leading Principal Minors . . . 89
6.17 Sylvester's Criterion . . . 89
6.18 Positive Semidefiniteness . . . 90
6.19 Cholesky Factorization and Positive Definiteness . . . 90
6.20 Quadratic Forms and Optimization . . . 91
6.21 Quadratic Forms and Second Derivatives . . . 91
6.22 Quadratic Hypersurfaces . . . 92
6.23 Congruence versus Orthogonal Diagonalization . . . 92
6.24 Summary of Real Quadratic Forms . . . 93

7 Singular Value Decomposition . . . 94
7.1 Why Singular Values? . . . 95
7.2 The Matrices A^*A and AA^* . . . 95
7.3 Singular Values . . . 96
7.4 Right Singular Vectors . . . 96
7.5 Left Singular Vectors . . . 97
7.6 The Singular Value Decomposition . . . 97
7.7 Compact Singular Value Decomposition . . . 99
7.8 Rank and Singular Values . . . 99
7.9 The Four Fundamental Subspaces . . . 100
7.10 Spectral Decomposition of A^*A . . . 100
7.11 Geometric Interpretation . . . 101
7.12 Images of the Unit Sphere . . . 101
7.13 Largest and Smallest Singular Values . . . 102
7.14 Operator Norm . . . 102
7.15 Frobenius Norm and Singular Values . . . 103
7.16 Rank-One Expansion . . . 103
7.17 Low-Rank Approximation . . . 103
7.18 The Moore–Penrose Pseudoinverse . . . 104
7.19 Least Squares via the SVD . . . 104
7.20 Singular Values and Determinants . . . 105
7.21 SVD versus Eigenvalue Decomposition . . . 105
7.22 SVD of a Normal Matrix . . . 106
7.23 Polar Decomposition . . . 106
7.24 Geometric Meaning of Polar Decomposition . . . 107
7.25 SVD as a Culmination of Spectral Theory . . . 107

8 Matrix Decompositions . . . 109
8.1 Why Matrix Decompositions? . . . 110
8.2 Several Different Types of Decomposition . . . 110
8.3 LU Decomposition . . . 111
8.4 LU Decomposition from Elimination . . . 111
8.5 Example of LU Decomposition . . . 111
8.6 Pivoting and the $PA = LU$ Form . . . 112
8.7 Solving Linear Systems with LU . . . 112
8.8 Rank Factorization . . . 112
8.9 Meaning of Rank Factorization . . . 113
8.10 QR Decomposition Revisited . . . 113

8.11 QR and Least Squares . . . 114
8.12 Cholesky Decomposition Revisited . . . 114
8.13 Cholesky and Quadratic Forms . . . 115
8.14 Schur Decomposition Revisited . . . 115
8.15 Eigendecomposition . . . 115
8.16 Jordan Decomposition Revisited . . . 116
8.17 SVD Revisited . . . 116
8.18 Polar Decomposition Revisited . . . 116
8.19 Relationships among Decompositions . . . 117
8.20 A Hierarchy of Structural Assumptions . . . 117
8.21 Which Decomposition Should Be Used? . . . 118
8.22 Canonical Forms versus Computational Factorizations . . . 118
8.23 Summary Table of Matrix Decompositions . . . 118
8.24 The Structural Thread of the Second Volume . . . 118
8.25 Completion of Undergraduate Advanced Linear Algebra . . . 119

Acknowledgements **120**

Introduction

The first volume developed the basic language and structure of finite-dimensional linear algebra. Starting from systems of linear equations and matrices, we introduced vector spaces and linear maps, interpreted matrices as coordinate representations of linear transformations, and studied quantities that remain invariant under changes of basis. Determinants, characteristic polynomials, and eigenvalues eventually led to diagonalization.
The central idea was that a complicated matrix may become much simpler after an appropriate change of coordinates. For a linear operator $T : V \to V$, changing a basis transforms its matrix by similarity:

$$A \longmapsto P^{-1}AP.$$

If V admits a basis consisting of eigenvectors of T, then this process reduces the matrix to a diagonal form

$$P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Diagonalization is powerful precisely because it separates the action of an operator into independent one-dimensional directions. However, not every operator possesses enough eigenvectors to admit such a basis. This limitation marks the natural starting point of the second volume.

0.1 Beyond Diagonalization

Suppose λ is an eigenvalue of T. The first volume established the inequality

$$1 \leq \dim E_\lambda(T) \leq m_a(\lambda),$$

where $m_a(\lambda)$ denotes the algebraic multiplicity of λ.
If

$$\dim E_\lambda(T) < m_a(\lambda)$$

for some eigenvalue, then ordinary eigenvectors cannot provide a basis of the entire space. Consequently, the operator is not diagonalizable.
This failure does not mean that the operator has no useful structure. Instead, ordinary eigenvectors must be supplemented by generalized eigenvectors. These vectors form Jordan chains and lead to a more general canonical representation: the Jordan normal form. Thus the first transition of this volume is

$$\boxed{\text{diagonalization} \longrightarrow \text{generalized eigenvectors} \longrightarrow \text{Jordan normal form.}}$$

Diagonal matrices are therefore not replaced by an unrelated theory. Rather, they appear as the simplest special case of Jordan normal form, in which every Jordan block has size one.

0.2 From Algebraic Structure to Geometric Structure

The vector spaces studied in the first volume carried only the operations of vector addition and scalar multiplication. Such a vector space has no intrinsic notion of length, angle, distance, or orthogonality.
An inner product introduces this additional geometric structure. For vectors u, v, the quantity

$$\langle u, v \rangle$$

allows us to define orthogonality, while

$$\|v\| = \sqrt{\langle v, v \rangle}$$

defines a norm.
Once an inner product is available, several familiar geometric ideas become part of linear algebra. Orthogonal complements provide natural decompositions of vector spaces, and the Gram–Schmidt process converts linearly independent vectors into an orthonormal basis.
This leads to a second structural direction:

vector spaces $\longrightarrow$ inner product spaces $\longrightarrow$ orthogonality $\longrightarrow$ orthonormal bases.

The allowed changes of coordinates also become more restrictive. Instead of using arbitrary invertible matrices, we may require the change of basis to preserve the inner product. This leads to orthogonal matrices over $\mathbb{R}$ and unitary matrices over $\mathbb{C}$.

0.3 Spectral Structure

The interaction between eigenvalue theory and inner products produces some of the strongest results in finite-dimensional linear algebra.
A general matrix need not be diagonalizable. Even when it is diagonalizable, its eigenvectors need not be orthogonal. Symmetric and Hermitian matrices behave much better. For a real symmetric matrix

$$A = A^T,$$

one can find an orthogonal matrix Q such that

$$Q^T A Q = D,$$

where D is diagonal.
Similarly, for a complex Hermitian matrix

$$A = A^*,$$

there exists a unitary matrix U such that

$$U^* A U = D.$$

Thus the appropriate geometric structure produces a stronger form of diagonalization:

symmetric or Hermitian structure $\longrightarrow$ orthonormal eigenbasis $\longrightarrow$ orthogonal or unitary diagonalizati

This spectral viewpoint forms one of the central themes of the second volume.

0.4 Bilinear and Quadratic Forms

Matrices can represent objects other than linear operators.
A bilinear form on a real vector space may be represented in coordinates as

$$B(x, y) = x^T A y,$$

while a quadratic form has the form

$$q(x) = x^T A x.$$

Here the natural change of coordinates is different from similarity. If

$$x = Py,$$

then

$$q(x) = y^T(P^TAP)y.$$

Thus the representing matrix transforms according to

$$A \longmapsto P^TAP.$$

This relation is called congruence.
The distinction is fundamental:

linear operator	quadratic form
$A \longmapsto P^{-1}AP$	$A \longmapsto P^TAP.$

Consequently, similarity and congruence solve different classification problems.
The study of quadratic forms leads naturally to positive and negative definiteness, canonical forms, inertia, and Sylvester's law of inertia. These results describe which properties of a quadratic form remain unchanged under arbitrary invertible changes of coordinates.

0.5 Matrix Decompositions

Canonical forms are not the only way to simplify matrices. Another major theme is to express a matrix as a product of matrices with simpler structures.
The singular value decomposition is one of the most important such factorizations. For a real matrix

$$A \in \mathbb{R}^{m\times n},$$

there exist orthogonal matrices U and V such that

$$A = U\Sigma V^T,$$

where Σ is diagonal in the rectangular sense and its nonnegative diagonal entries are the singular values of A.
Unlike diagonalization, the singular value decomposition exists for every matrix, including rectangular matrices and matrices that are not diagonalizable.
Geometrically, the decomposition separates a linear transformation into three stages:

rotation or reflection $\longrightarrow$ independent scaling $\longrightarrow$ rotation or reflection.

The SVD therefore connects matrix algebra, inner product geometry, eigenvalue theory, and computation.

0.6 Different Ways to Simplify a Matrix

A central lesson of advanced linear algebra is that there is no single notion of a canonical matrix. The appropriate simplification depends on what mathematical structure the matrix represents.
For a linear operator, a change of basis produces similarity:

$$A \longmapsto P^{-1}AP.$$

This leads to diagonal and Jordan forms.
For a quadratic form, a change of variables produces congruence:

$$A \longmapsto P^T AP.$$

This leads to canonical forms and the theory of inertia.
When an inner product must be preserved, the change of basis is restricted to orthogonal or unitary transformations.
For a general rectangular matrix, the domain and codomain may be changed independently, leading to the singular value decomposition

$$A = U\Sigma V^*.$$

These constructions are unified by one principle:

identify the structure $\longrightarrow$ choose the correct equivalence $\longrightarrow$ find the simplest representative.

0.7 Scope of the Second Volume

The purpose of this volume is to complete the core finite-dimensional theory of undergraduate advanced linear algebra.
Its subject is therefore broader than elementary matrix computation, but it deliberately stops before the language of later abstract algebra and geometry becomes essential.
Topics such as modules and quotient modules, exact sequences, tensor products, and more advanced multilinear constructions are not part of the present volume. Likewise, structures leading toward symplectic geometry, Hamiltonian systems, Lie algebras, and commutative algebra are separated from the undergraduate linear algebra sequence.
Such material is collected instead in the companion volume *Algebraic Frontiers*.
The three volumes therefore follow the broad progression

Advanced Linear Algebra I foundations, linear operators, and diagonalization

$$\Downarrow$$

Advanced Linear Algebra II canonical forms, inner products, forms, and decompositions

while more abstract extensions are separated into

Algebraic Frontiers connections with later algebraic theories.

0.8 Organization of This Volume

The development of this volume follows four closely related themes.
First, the failure of diagonalization leads to generalized eigenvectors and Jordan normal form:

$$\text{eigenvectors} \longrightarrow \text{generalized eigenvectors} \longrightarrow \text{Jordan blocks}.$$

Second, inner products introduce geometry:

$$\text{inner products} \longrightarrow \text{orthogonality} \longrightarrow \text{Gram–Schmidt} \longrightarrow \text{orthonormal bases}.$$

Third, bilinear and quadratic forms introduce a different classification problem:

$$\text{bilinear forms} \longrightarrow \text{quadratic forms} \longrightarrow \text{congruence} \longrightarrow \text{canonical forms}.$$

Finally, matrix decompositions combine algebraic and geometric structure:

$$\text{orthogonality} \longrightarrow \text{spectral theory} \longrightarrow \text{singular value decomposition}.$$

The unifying objective is not merely to calculate with matrices, but to understand which aspects of a linear object are intrinsic and which arise only from the coordinates used to represent it.
The first volume established this distinction. The second develops it further by showing that different structures require different notions of equivalence and therefore different canonical forms.
In this sense, the main thread running through the two volumes is

$$\boxed{\text{linear structure} \longrightarrow \text{invariants} \longrightarrow \text{canonical forms} \longrightarrow \text{decomposition}.}$$

Chapter 1

Jordan Normal Form

1.1 From Diagonalization to Jordan Form

Let V be a finite-dimensional vector space over a field K, and let

$$T \in \mathrm{End}_K(V).$$

In the first volume, we studied diagonalization. If V has a basis consisting of eigenvectors of T, then

$$[T]_{\mathcal{B}} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Equivalently, a matrix $A \in M_n(K)$ is diagonalizable if there exists an invertible matrix P such that

$$P^{-1}AP = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

However, not every matrix is diagonalizable.
Consider

$$A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

Its characteristic polynomial is

$$\chi_A(X) = (X - \lambda)^2.$$

But

$$A - \lambda I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

so

$$\dim \ker(A - \lambda I) = 1.$$

Thus the algebraic multiplicity of λ is 2, while its geometric multiplicity is only 1. There are not enough eigenvectors to form a basis.
Nevertheless, the matrix is already extremely simple.
This motivates the central problem of this chapter:

If a matrix cannot be diagonalized, how close to diagonal form can it be made?

The answer is given by Jordan normal form.

1.2 Polynomial Relations and Annihilating Polynomials

Let

$$T \in \mathrm{End}_K(V)$$

and let

$$p(X) = a_0 + a_1X + \cdots + a_mX^m \in K[X].$$

Recall that

$$p(T) = a_0I + a_1T + \cdots + a_mT^m.$$

Definition 1.2.1. A polynomial $p \in K[X]$ is called an annihilating polynomial of T if

$$p(T) = 0.$$

By the Cayley–Hamilton theorem,

$$\chi_T(T) = 0.$$

Therefore the characteristic polynomial is always an annihilating polynomial.
If

$$p(T) = 0$$

and λ is an eigenvalue of T, then

$$p(\lambda) = 0.$$

Indeed, if

$$T(v) = \lambda v$$

for some $v \neq 0$, then

$$p(T)v = p(\lambda)v.$$

Since $p(T) = 0$,

$$p(\lambda)v = 0,$$

and therefore

$$p(\lambda) = 0.$$

Thus every eigenvalue must be a root of every annihilating polynomial.

1.3 The Minimal Polynomial

Among all polynomial relations satisfied by T, there is a unique one of smallest degree after normalization.

Definition 1.3.1. The minimal polynomial of T, denoted by

$$m_T(X),$$

is the unique monic polynomial of smallest positive degree satisfying

$$m_T(T) = 0.$$

For a matrix A, we similarly write

$$m_A(X).$$

Proposition 1.3.2. If $p(T) = 0$, then

$$m_T(X) \mid p(X).$$

Proof. By the division algorithm, write

$$p(X) = q(X)m_T(X) + r(X),$$

where either $r = 0$ or

$$\deg r < \deg m_T.$$

Evaluating at T,

$$0 = p(T) = q(T)m_T(T) + r(T) = r(T).$$

If $r \neq 0$, this contradicts the minimality of $\deg m_T$. Hence $r = 0$, and therefore

$$m_T \mid p.$$

□

Since

$$\chi_T(T) = 0,$$

we immediately obtain

$$\boxed{m_T(X) \mid \chi_T(X).}$$

Thus the minimal polynomial contains less redundant information than the characteristic polynomial while still recording a polynomial relation satisfied by the operator.

1.4 Minimal Polynomial and Similarity

Theorem 1.4.1. Similar matrices have the same minimal polynomial.

Proof. Suppose

$$B = P^{-1}AP.$$

For every polynomial p,

$$p(B) = P^{-1}p(A)P.$$

Hence

$$p(A) = 0 \iff p(B) = 0.$$

Therefore A and B have exactly the same annihilating polynomials, so

$$m_A(X) = m_B(X).$$

□

Consequently, the minimal polynomial is a similarity invariant.

1.5 A Diagonalization Criterion

The minimal polynomial gives a concise characterization of diagonalizability.

Theorem 1.5.1. Let $T \in \mathrm{End}_K(V)$. Then T is diagonalizable over K if and only if

$$m_T(X)$$

splits over K into distinct linear factors.

In other words,

$$T \text{ is diagonalizable}$$

if and only if

$$m_T(X) = (X - \lambda_1) \cdots (X - \lambda_r),$$

where the λ_i are distinct.

Proof. Suppose first that T is diagonalizable. Choose an eigenbasis. Then

$$[T] = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

If the distinct eigenvalues are

$$\mu_1, \ldots, \mu_r,$$

then

$$p(X) = \prod_{i=1}^{r} (X - \mu_i)$$

satisfies

$$p(T) = 0.$$

Hence

$$m_T \mid p.$$

Since every eigenvalue must be a root of m_T, we obtain

$$m_T(X) = \prod_{i=1}^{r} (X - \mu_i).$$

Conversely, suppose

$$m_T(X) = \prod_{i=1}^{r} (X - \lambda_i)$$

with distinct λ_i. Then the polynomial factors are pairwise coprime. The corresponding eigenspaces decompose V as

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_r}(T).$$

Choosing a basis in each eigenspace gives an eigenbasis of V. Therefore T is diagonalizable. □

The failure of diagonalization is therefore reflected by repeated factors in the minimal polynomial.

1.6 Generalized Eigenvectors

Ordinary eigenvectors satisfy

$$(T - \lambda I)v = 0.$$

When the eigenspace is too small, we enlarge it by allowing higher powers of $T - \lambda I$.

Definition 1.6.1. A nonzero vector $v \in V$ is called a generalized eigenvector associated with the eigenvalue λ if

$$(T - \lambda I)^k v = 0$$

for some positive integer k.

Every ordinary eigenvector is therefore a generalized eigenvector with $k = 1$.

Definition 1.6.2. For $k \geq 1$, define

$$G_\lambda^{(k)} = \ker(T - \lambda I)^k.$$

These spaces form an increasing sequence:

$$\ker(T - \lambda I) \subseteq \ker(T - \lambda I)^2 \subseteq \ker(T - \lambda I)^3 \subseteq \cdots .$$

Since V is finite-dimensional, this sequence eventually stabilizes.

Definition 1.6.3. The generalized eigenspace associated with λ is

$$G_\lambda(T) = \ker(T - \lambda I)^N$$

for any sufficiently large N.

In particular, one may always take

$$N = \dim V.$$

Thus

$$E_\lambda(T) \subseteq G_\lambda(T).$$

The difference between these two spaces measures precisely the additional vectors needed when ordinary eigenvectors are insufficient.

1.7 Stabilization of Kernels

Let

$$N = T - \lambda I.$$

Then

$$\ker N \subseteq \ker N^2 \subseteq \cdots .$$

Proposition 1.7.1. If

$$\ker N^k = \ker N^{k+1}$$

for some $k \geq 1$, then

$$\ker N^k = \ker N^{k+r}$$

for every $r \geq 0$.

Proof. It suffices to show that

$$\ker N^{k+1} = \ker N^{k+2}.$$

Let

$$v \in \ker N^{k+2}.$$

Then

$$N^{k+2}v = 0,$$

so

$$N^{k+1}(Nv) = 0.$$

Thus

$$Nv \in \ker N^{k+1} = \ker N^k.$$

Hence

$$N^k(Nv) = 0,$$

which gives

$$N^{k+1}v = 0.$$

Therefore

$$v \in \ker N^{k+1}.$$

The reverse inclusion is automatic. □

Since dimensions cannot increase indefinitely, stabilization must occur after finitely many steps.

1.8 Primary Decomposition

Suppose the characteristic polynomial of T splits over K:

$$\chi_T(X) = \prod_{i=1}^{r} (X - \lambda_i)^{m_i},$$

where the λ_i are distinct.
For each eigenvalue, define

$$G_{\lambda_i}(T) = \ker(T - \lambda_i I)^{m_i}.$$

Theorem 1.8.1 (Primary Decomposition)**.** We have

$$\boxed{V = G_{\lambda_1}(T) \oplus \cdots \oplus G_{\lambda_r}(T).}$$

Moreover, every generalized eigenspace

$$G_{\lambda_i}(T)$$

is T-invariant.
Thus the study of a general operator whose characteristic polynomial splits reduces to studying its restriction to each generalized eigenspace.
On $G_\lambda(T)$, the operator has the form

$$T = \lambda I + N,$$

where

$$N = T - \lambda I$$

is nilpotent.
Therefore the Jordan problem reduces to understanding nilpotent operators.

1.9 Nilpotent Operators

Definition 1.9.1. A linear operator $N : V \to V$ is nilpotent if there exists a positive integer k such that

$$N^k = 0.$$

The smallest such k is called the nilpotency index of N.
If

$$N^k = 0,$$

then every eigenvalue of N is zero.
Indeed, if

$$Nv = \lambda v,$$

then

$$0 = N^k v = \lambda^k v.$$

Since $v \neq 0$,

$$\lambda = 0.$$

Thus

$$\chi_N(X) = X^n.$$

Similarly, the minimal polynomial of a nilpotent operator has the form

$$m_N(X) = X^s$$

for some $1 \leq s \leq n$.

1.10 Jordan Chains

The structure of a nilpotent operator is described by chains of vectors.

Definition 1.10.1. A sequence of nonzero vectors

$$v_1, v_2, \ldots, v_k$$

is called a Jordan chain for T associated with the eigenvalue λ if

$$(T - \lambda I)v_1 = 0$$

and

$$(T - \lambda I)v_j = v_{j-1}, \qquad j = 2, \ldots, k.$$

Thus

$$v_1$$

is an ordinary eigenvector, while

$$v_2, \ldots, v_k$$

are generalized eigenvectors.
The chain has the form

$$v_k \longmapsto v_{k-1} \longmapsto \cdots \longmapsto v_2 \longmapsto v_1 \longmapsto 0$$

under $T - \lambda I$.
Equivalently,

$$(T - \lambda I)^j v_k = v_{k-j}$$

for $j < k$, while

$$(T - \lambda I)^k v_k = 0.$$

Proposition 1.10.2. Vectors in a Jordan chain are linearly independent.

Proof. Suppose

$$a_1 v_1 + \cdots + a_k v_k = 0.$$

Applying

$$(T - \lambda I)^{k-1}$$

gives

$$a_k v_1 = 0.$$

Hence

$$a_k = 0.$$

Repeating the argument successively gives

$$a_{k-1} = \cdots = a_1 = 0.$$

Therefore the chain vectors are linearly independent. □

1.11 Jordan Blocks

Definition 1.11.1. The $k \times k$ Jordan block associated with λ is

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}.$$

We may write

$$J_k(\lambda) = \lambda I_k + N_k,$$

where

$$N_k = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{pmatrix}.$$

The matrix N_k is nilpotent and satisfies

$$N_k^k = 0, \qquad N_k^{k-1} \neq 0.$$

Hence

$$m_{J_k(\lambda)}(X) = (X - \lambda)^k.$$

Also,

$$\chi_{J_k(\lambda)}(X) = (X - \lambda)^k.$$

1.12 Jordan Chains and Jordan Blocks

Suppose

$$v_1, \ldots, v_k$$

is a Jordan chain satisfying

$$(T - \lambda I)v_1 = 0$$

and

$$(T - \lambda I)v_j = v_{j-1}.$$

If we choose the ordered basis

$$\mathcal{B} = (v_1, \ldots, v_k),$$

then

$$T(v_1) = \lambda v_1$$

and

$$T(v_j) = v_{j-1} + \lambda v_j.$$

Depending on the ordering convention for the chain, the corresponding Jordan block may have its 1's on the superdiagonal or subdiagonal.

Throughout these notes, Jordan blocks are written with 1's on the superdiagonal:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix}.$$

Accordingly, the basis vectors may be ordered in the direction compatible with this convention.

Remark 1.12.1. Different textbooks may reverse the order of the vectors in a Jordan chain. This changes whether the 1's appear above or below the diagonal, but the resulting matrices are similar and describe the same structure.

1.13 Jordan Normal Form

Definition 1.13.1. A matrix is in Jordan normal form if it is block diagonal and every diagonal block is a Jordan block:

$$J = \operatorname{diag}\Big(J_{k_1}(\lambda_1), \ldots, J_{k_r}(\lambda_r)\Big).$$

For example,

$$J = \begin{pmatrix} \lambda & 1 & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 & 0 \\ 0 & 0 & \lambda & 1 & 0 \\ 0 & 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & 0 & \mu \end{pmatrix}$$

contains two Jordan blocks associated with λ and one block associated with μ.

Theorem 1.13.2 (Jordan Normal Form)**.** Let $A \in M_n(K)$. If the characteristic polynomial of A splits completely over K, then there exists an invertible matrix P such that

$$\boxed{P^{-1}AP = J,}$$

where J is in Jordan normal form.

In particular, every complex matrix has a Jordan normal form because every polynomial over $\mathbb{C}$ splits completely.
Over $\mathbb{R}$, a real matrix need not possess a real Jordan normal form if its characteristic polynomial has nonreal roots.

1.14 The Structure of a Jordan Form

For each eigenvalue λ, the corresponding Jordan blocks describe the restriction of T to the generalized eigenspace

$$G_\lambda(T).$$

Suppose the blocks associated with λ have sizes

$$k_1, \ldots, k_s.$$

Then the λ-part of the Jordan form is

$$J_{k_1}(\lambda) \oplus \cdots \oplus J_{k_s}(\lambda).$$

Several important quantities can be read immediately from these blocks.
The algebraic multiplicity is

$$m_a(\lambda) = k_1 + \cdots + k_s.$$

The geometric multiplicity is

$$m_g(\lambda) = s.$$

Thus

$$\boxed{m_g(\lambda) = \text{number of Jordan blocks associated with } \lambda.}$$

The exponent of $X - \lambda$ in the minimal polynomial is

$$\max\{k_1, \ldots, k_s\}.$$

Therefore

$$\boxed{\text{largest Jordan block for } \lambda = \text{exponent of } X - \lambda \text{ in } m_T.}$$

1.15 Diagonalization Revisited

Jordan form immediately recovers the diagonalization criterion.
A Jordan block

$$J_k(\lambda)$$

is diagonal if and only if

$$k = 1.$$

Therefore:

Theorem 1.15.1. A matrix A is diagonalizable if and only if every Jordan block of A has size 1.

Equivalently,

$$A \text{ is diagonalizable} \iff m_A(X)$$

has no repeated linear factor.
For each eigenvalue λ,

$$m_g(\lambda) = m_a(\lambda)$$

if and only if every Jordan block associated with λ has size 1.
Thus ordinary diagonalization appears as the simplest possible Jordan structure.

1.16 Determining Jordan Blocks from Kernel Dimensions

Suppose λ is an eigenvalue and define

$$N = A - \lambda I.$$

Let

$$d_k = \dim \ker N^k, \qquad d_0 = 0.$$

The sequence

$$d_1, d_2, d_3, \ldots$$

contains complete information about the sizes of the Jordan blocks associated with λ.

Theorem 1.16.1. The number of Jordan blocks associated with λ having size at least k is

$$\boxed{d_k - d_{k-1}.}$$

Consequently, the number of blocks of size exactly k is

$$\boxed{2d_k - d_{k-1} - d_{k+1}.}$$

To see the first formula, consider a single Jordan block of size s. For this block,

$$\dim\ker N^k = \min\{k, s\}.$$

Therefore

$$\dim\ker N^k - \dim\ker N^{k-1} = \begin{cases} 1, & s \geq k, \\ 0, & s < k. \end{cases}$$

Summing over all Jordan blocks gives the result.

1.17 Example: Recovering the Jordan Structure

Suppose an eigenvalue λ has algebraic multiplicity 6, and suppose

$$\dim\ker(A - \lambda I) = 3,$$

$$\dim\ker(A - \lambda I)^2 = 5,$$

and

$$\dim\ker(A - \lambda I)^3 = 6.$$

Set

$$d_0 = 0, \qquad d_1 = 3, \qquad d_2 = 5, \qquad d_3 = 6.$$

The number of Jordan blocks is

$$d_1 - d_0 = 3.$$

The number of blocks of size at least 2 is

$$d_2 - d_1 = 2.$$

The number of blocks of size at least 3 is

$$d_3 - d_2 = 1.$$

Therefore the block sizes are

$$3, \quad 2, \quad 1.$$

Hence the λ-part of the Jordan form is

$$J_3(\lambda) \oplus J_2(\lambda) \oplus J_1(\lambda).$$

This method is usually more systematic than attempting to guess Jordan chains directly.

1.18 Constructing a Jordan Basis

Knowing the sizes of the Jordan blocks determines the Jordan form, but constructing the change-of-basis matrix P requires actual Jordan chains.
Suppose

$$N = A - \lambda I.$$

To construct a Jordan chain of length k, choose

$$v_k \in \ker N^k$$

such that

$$v_k \notin \ker N^{k-1}.$$

Then define

$$v_{k-1} = N v_k,$$
$$v_{k-2} = N^2 v_k,$$

and continue until

$$v_1 = N^{k-1} v_k.$$

Then

$$N v_1 = 0,$$

and

$$N v_j = v_{j-1}.$$

Hence

$$v_1, \ldots, v_k$$

form a Jordan chain.
Repeating this construction with sufficiently many independent chains produces a Jordan basis of the generalized eigenspace.
Finally, combining Jordan bases for all distinct eigenvalues gives a Jordan basis of V.

1.19 Characteristic and Minimal Polynomials from Jordan Form

Suppose

$$J = \operatorname{diag}\Big(J_{k_1}(\lambda_1), \ldots, J_{k_r}(\lambda_r)\Big).$$

The characteristic polynomial is obtained by multiplying the characteristic polynomials of all blocks:

$$\chi_J(X) = \prod_{i=1}^{r} (X - \lambda_i)^{k_i}.$$

To determine the minimal polynomial, group blocks according to their eigenvalues.
Suppose the distinct eigenvalues are

$$\mu_1, \ldots, \mu_s,$$

and let r_i be the size of the largest Jordan block associated with μ_i. Then

$$\boxed{m_J(X) = \prod_{i=1}^{s} (X - \mu_i)^{r_i}.}$$

Thus the characteristic polynomial records the total size of all blocks, while the minimal polynomial records the largest block for each eigenvalue.

1.20 Uniqueness of Jordan Normal Form

The order of Jordan blocks along the diagonal is irrelevant.
For example,

$$J_2(\lambda) \oplus J_1(\mu)$$

and

$$J_1(\mu) \oplus J_2(\lambda)$$

represent the same Jordan structure and are similar.
Apart from this reordering, the Jordan form is unique.

Theorem 1.20.1 (Uniqueness of Jordan Normal Form)**.** Suppose the characteristic polynomial of A splits over K. Then the Jordan normal form of A is unique up to permutation of its Jordan blocks.

Indeed, for each eigenvalue λ, the dimensions

$$\dim \ker(A - \lambda I)^k, \qquad k = 1, 2, \ldots,$$

are similarity invariants.
These dimensions determine the number of blocks of every possible size. Hence the complete collection of Jordan blocks is uniquely determined.

1.21 Similarity Classification by Jordan Form

Jordan normal form solves the similarity classification problem over a field where the relevant characteristic polynomials split.

Theorem 1.21.1. Let $A, B \in M_n(K)$, and suppose their characteristic polynomials split over K. Then

$$A \sim B$$

if and only if A and B have the same Jordan normal form up to reordering of Jordan blocks.

Thus the Jordan form is not merely a convenient simplification. It is a complete invariant for similarity.
The characteristic polynomial alone is not sufficient.
For example,

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

and

$$B = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

have the same characteristic polynomial

$$(X - \lambda)^2.$$

However,

$$m_A(X) = X - \lambda,$$

while

$$m_B(X) = (X - \lambda)^2.$$

Hence they are not similar.
Even the characteristic polynomial together with the minimal polynomial may fail to determine the entire Jordan form in higher dimensions. The full sequence

$$\dim \ker(A - \lambda I)^k$$

contains the necessary information.

1.22 Functions and Powers of Jordan Blocks

Jordan form is useful not only for classification but also for computation.
Write

$$J_k(\lambda) = \lambda I + N,$$

where $N^k = 0$.
Since λI commutes with N, the binomial theorem gives

$$J_k(\lambda)^m = (\lambda I + N)^m = \sum_{j=0}^{k-1} \binom{m}{j} \lambda^{m-j} N^j.$$

The sum stops at $k-1$ because

$$N^k = 0.$$

For example,

$$J_2(\lambda)^m = \begin{pmatrix} \lambda^m & m\lambda^{m-1} \\ 0 & \lambda^m \end{pmatrix}.$$

Similarly,

$$J_3(\lambda)^m = \begin{pmatrix} \lambda^m & m\lambda^{m-1} & \binom{m}{2}\lambda^{m-2} \\ 0 & \lambda^m & m\lambda^{m-1} \\ 0 & 0 & \lambda^m \end{pmatrix}.$$

Thus Jordan form makes high powers of a matrix explicitly computable.

1.23 The Exponential of a Jordan Block

The same principle applies to the matrix exponential.
For

$$J = \lambda I + N,$$

we have

$$e^{tJ} = e^{t\lambda I} e^{tN} = e^{\lambda t} e^{tN}.$$

Since N is nilpotent,

$$e^{tN} = I + tN + \frac{t^2}{2!}N^2 + \cdots + \frac{t^{k-1}}{(k-1)!}N^{k-1}.$$

Hence

$$\boxed{e^{tJ_k(\lambda)} = e^{\lambda t}\left(I + tN + \cdots + \frac{t^{k-1}}{(k-1)!}N^{k-1}\right).}$$

For example,

$$e^{tJ_3(\lambda)} = e^{\lambda t}\begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

This explains why solutions of linear differential systems associated with non-diagonalizable matrices contain terms such as

$$t^j e^{\lambda t}.$$

Thus Jordan form also provides a direct connection between linear algebra and systems of ordinary differential equations.

1.24 The Structural Meaning of Jordan Form

The Jordan normal form separates a linear operator into two basic components.
On a generalized eigenspace associated with λ,

$$T = \lambda I + N,$$

where N is nilpotent.
The scalar part

$$\lambda I$$

describes the eigenvalue, while the nilpotent part N measures the failure of diagonalizability.
For a Jordan block,

$$J_k(\lambda) = \lambda I_k + N_k.$$

If

$$N_k = 0,$$

then $k = 1$, and the block is diagonal.
Larger nilpotent parts correspond to longer Jordan chains and hence to a greater failure of ordinary eigenvectors to span the space.
Thus Jordan theory refines the eigenvalue theory of the first volume:

eigenvalues $\longrightarrow$ eigenspaces $\longrightarrow$ generalized eigenspaces $\longrightarrow$ Jordan chains $\longrightarrow$ Jordan normal form.

The characteristic polynomial determines the eigenvalues and their total algebraic multiplicities. The minimal polynomial determines the largest Jordan block associated with each eigenvalue. The dimensions

$$\dim \ker (T - \lambda I)^k$$

determine the complete collection of Jordan blocks.
Consequently, Jordan normal form gives a complete description of a linear operator up to similarity whenever its characteristic polynomial splits.
Having completed the similarity theory of general linear operators, we now turn to vector spaces equipped with additional geometric structure. The next chapter introduces inner products, orthogonality, orthogonal complements, and the Gram–Schmidt process.

Chapter 2

Inner Product Spaces and Gram–Schmidt Orthogonalization

2.1 Inner Products

Let V be a finite-dimensional vector space over

$$K = \mathbb{R} \qquad \text{or} \qquad K = \mathbb{C}.$$

An inner product equips V with geometric notions such as length, angle, and orthogonality.

Definition 2.1.1. An inner product on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \to K$$

satisfying the following properties for all $x, y, z \in V$ and $a, b \in K$:

1. linearity in the first variable:

$$\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle ;$$

2. conjugate symmetry:

$$\langle x, y \rangle = \overline{\langle y, x \rangle};$$

3. positive definiteness:

$$\langle x, x \rangle \geq 0,$$

and

$$\langle x, x \rangle = 0 \iff x = 0.$$

Over $\mathbb{R}$, complex conjugation disappears, so conjugate symmetry reduces to

$$\langle x, y \rangle = \langle y, x \rangle .$$

Over $\mathbb{C}$, conjugate symmetry implies conjugate linearity in the second variable:

$$\langle x, ay + bz \rangle = \overline{a} \langle x, y \rangle + \overline{b} \langle x, z \rangle .$$

Remark 2.1.2. Some textbooks choose the opposite convention and take the inner product to be linear in the second variable instead. The two conventions are equivalent if used consistently.

Definition 2.1.3. A vector space equipped with an inner product is called an inner product space.

2.2 Standard Inner Products

The standard inner product on $\mathbb{R}^n$ is

$$\langle x, y \rangle = x^T y = \sum_{i=1}^{n} x_i y_i.$$

For $\mathbb{C}^n$, the standard inner product is

$$\langle x, y \rangle = x^* y = \sum_{i=1}^{n} x_i \overline{y_i},$$

where

$$x^* = \overline{x}^T.$$

Thus

$$\langle x, x \rangle = \sum_{i=1}^{n} |x_i|^2 \geq 0.$$

Example 2.2.1. For

$$x = \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad y = \begin{pmatrix} 2 \\ 1 - i \end{pmatrix},$$

we have

$$\langle x, y \rangle = 1 \cdot 2 + i(1 + i) = 1 + i.$$

More generally, if

$$A \in M_n(\mathbb{R})$$

is symmetric positive definite, then

$$\langle x, y \rangle_A = x^T A y$$

defines an inner product on $\mathbb{R}^n$.
Similarly, if

$$A \in M_n(\mathbb{C})$$

is Hermitian positive definite, then

$$\langle x, y \rangle_A = x^* A y$$

defines an inner product on $\mathbb{C}^n$.

2.3 Norms

An inner product determines a notion of length.

Definition 2.3.1. The norm induced by an inner product is

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

The norm satisfies

$$\|x\| \geq 0,$$

$$\|x\| = 0 \iff x = 0,$$

and

$$\|ax\| = |a| \, \|x\| \, .$$

For the standard inner product on K^n,

$$\|x\| = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2}.$$

Definition 2.3.2. A vector x is called a unit vector if

$$\|x\| = 1.$$

If $x \neq 0$, then

$$\frac{x}{\|x\|}$$

is a unit vector in the same direction as x.
This process is called normalization.

2.4 Cauchy–Schwarz Inequality

One of the fundamental inequalities in inner product spaces is the Cauchy–Schwarz inequality.

Theorem 2.4.1 (Cauchy–Schwarz Inequality)**.** For all $x, y \in V$,

$$|\langle x, y\rangle| \leq \|x\| \, \|y\| .$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $y = 0$, the result is immediate.
Suppose $y \neq 0$. For

$$a = \frac{\langle x, y\rangle}{\langle y, y\rangle},$$

positive definiteness gives

$$0 \leq \|x - ay\|^2 .$$

Expanding,

$$\|x - ay\|^2 = \langle x - ay, x - ay\rangle .$$

Using sesquilinearity,

$$\|x - ay\|^2 = \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}.$$

Therefore

$$|\langle x, y\rangle|^2 \leq \|x\|^2 \, \|y\|^2 .$$

Taking square roots gives

$$|\langle x, y\rangle| \leq \|x\| \, \|y\| .$$

□

2.5 Triangle Inequality

The Cauchy–Schwarz inequality implies the triangle inequality.

Theorem 2.5.1 (Triangle Inequality)**.** For all $x, y \in V$,

$$\|x + y\| \leq \|x\| + \|y\| .$$

Proof. We have

$$\|x + y\|^2 = \langle x + y, x + y\rangle .$$

Hence

$$\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re} \langle x, y\rangle + \|y\|^2 .$$

By Cauchy–Schwarz,

$$\operatorname{Re} \langle x, y\rangle \leq |\langle x, y\rangle| \leq \|x\| \, \|y\| .$$

Therefore

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2 .$$

Taking square roots yields the result. □

Thus every inner product defines a metric by

$$d(x, y) = \|x - y\| .$$

2.6 Angles

In a real inner product space, the angle between nonzero vectors x and y may be defined through

$$\cos\theta = \frac{\langle x, y\rangle}{\|x\|\,\|y\|}.$$

The Cauchy–Schwarz inequality guarantees that

$$-1 \le \frac{\langle x, y\rangle}{\|x\|\,\|y\|} \le 1.$$

Hence

$$0 \le \theta \le \pi.$$

In complex inner product spaces, the inner product may be complex, so the notion of angle is less direct. Orthogonality, however, remains perfectly well defined.

2.7 Orthogonality

Definition 2.7.1. Vectors $x, y \in V$ are orthogonal if

$$\langle x, y\rangle = 0.$$

We write

$$x \perp y.$$

A collection of vectors

$$v_1, \dots, v_r$$

is orthogonal if

$$\langle v_i, v_j\rangle = 0$$

whenever $i \neq j$.
It is orthonormal if, in addition,

$$\|v_i\| = 1$$

for every i.
Thus an orthonormal collection satisfies

$$\langle v_i, v_j\rangle = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Proposition 2.7.2. Every orthogonal collection of nonzero vectors is linearly independent.

Proof. Suppose

$$a_1 v_1 + \cdots + a_r v_r = 0.$$

Taking the inner product with v_j,

$$0 = \langle a_1 v_1 + \cdots + a_r v_r, v_j\rangle\,.$$

By orthogonality,

$$0 = a_j \|v_j\|^2\,.$$

Since $v_j \neq 0$,

$$a_j = 0.$$

This holds for every j. □

2.8 Pythagorean Theorem

Orthogonality generalizes the familiar Pythagorean theorem.

Theorem 2.8.1. If

$$x \perp y,$$

then

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 .$$

More generally, if

$$v_1, \dots, v_r$$

are pairwise orthogonal, then

$$\|v_1 + \cdots + v_r\|^2 = \sum_{i=1}^{r} \|v_i\|^2 .$$

For orthonormal vectors,

$$\|a_1 v_1 + \cdots + a_r v_r\|^2 = \sum_{i=1}^{r} |a_i|^2 .$$

2.9 Orthonormal Bases

Definition 2.9.1. An orthonormal basis of V is a basis

$$\mathcal{B} = (e_1, \dots, e_n)$$

such that

$$\langle e_i, e_j \rangle = \delta_{ij}.$$

Orthonormal bases simplify coordinate calculations dramatically.
Suppose

$$x = a_1 e_1 + \cdots + a_n e_n.$$

Taking the inner product with e_j,

$$\langle x, e_j \rangle = a_j.$$

Therefore

$$\boxed{x = \sum_{j=1}^{n} \langle x, e_j \rangle \, e_j.}$$

Hence the coordinates of x in an orthonormal basis are obtained directly by inner products:

$$[x]_{\mathcal{B}} = \begin{pmatrix} \langle x, e_1 \rangle \\ \vdots \\ \langle x, e_n \rangle \end{pmatrix}.$$

Moreover,

$$\|x\|^2 = \sum_{j=1}^{n} |\langle x, e_j \rangle|^2 .$$

This is the finite-dimensional Parseval identity.

2.10 Gram Matrices

Let

$$v_1, \ldots, v_r \in V.$$

Definition 2.10.1. The Gram matrix associated with $v_1, \ldots, v_r$ is

$$G = \left(\langle v_i, v_j \rangle\right)_{i,j=1}^{r}.$$

Thus

$$G = \begin{pmatrix} \langle v_1, v_1 \rangle & \cdots & \langle v_1, v_r \rangle \\ \vdots & \ddots & \vdots \\ \langle v_r, v_1 \rangle & \cdots & \langle v_r, v_r \rangle \end{pmatrix}.$$

The Gram matrix is Hermitian:

$$G^* = G.$$

For any

$$a = \begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix},$$

we have

$$a^* G a = \left\| a_1 v_1 + \cdots + a_r v_r \right\|^2.$$

Therefore

$$a^* G a \geq 0.$$

Proposition 2.10.2. The Gram matrix G is invertible if and only if

$$v_1, \ldots, v_r$$

are linearly independent.

Proof. If the vectors are linearly dependent, there exists $a \neq 0$ such that

$$a_1 v_1 + \cdots + a_r v_r = 0.$$

Hence

$$a^* G a = 0,$$

and in fact $Ga = 0$, so G is singular.
Conversely, if G is singular, choose $a \neq 0$ with

$$Ga = 0.$$

Then

$$0 = a^* G a = \left\| \sum_{i=1}^{r} a_i v_i \right\|^2.$$

Hence

$$\sum_{i=1}^{r} a_i v_i = 0,$$

so the vectors are linearly dependent. □

For an orthonormal basis,

$$G = I.$$

2.11 Orthogonal Complements

Let $W \subseteq V$ be a subspace.

Definition 2.11.1. The orthogonal complement of W is

$$W^{\perp} = \{v \in V : \langle v, w \rangle = 0 \text{ for every } w \in W\}.$$

Proposition 2.11.2. The set $W^{\perp}$ is a subspace of V.

Also,

$$W \cap W^{\perp} = \{0\}.$$

In finite-dimensional inner product spaces, we have the fundamental orthogonal decomposition.

Theorem 2.11.3. Let $W \subseteq V$ be a subspace. Then

$$\boxed{V = W \oplus W^{\perp}.}$$

Consequently,

$$\dim V = \dim W + \dim W^{\perp}.$$

Therefore

$$\dim W^{\perp} = \dim V - \dim W.$$

Corollary 2.11.4. For every subspace W,

$$(W^{\perp})^{\perp} = W.$$

2.12 Orthogonal Projection onto a Subspace

The decomposition

$$V = W \oplus W^{\perp}$$

means that every $x \in V$ can be written uniquely as

$$x = w + z,$$

where

$$w \in W, \qquad z \in W^{\perp}.$$

Definition 2.12.1. The orthogonal projection of x onto W is

$$P_W x = w.$$

Thus

$$x - P_W x \in W^{\perp}.$$

The map

$$P_W : V \to V$$

is linear.
It satisfies

$$P_W^2 = P_W.$$

Moreover,

$$\operatorname{Im} P_W = W$$

and

$$\ker P_W = W^{\perp}.$$

Proposition 2.12.2. The orthogonal projection P_W is characterized by

$$P_W x \in W$$

and

$$x - P_W x \perp W.$$

This gives a geometric interpretation: $P_W x$ is the point of W closest to x.

2.13 Projection Formula

Suppose

$$e_1, \ldots, e_r$$

is an orthonormal basis of W.
Then

$$\boxed{P_W x = \sum_{i=1}^{r} \langle x, e_i \rangle \, e_i.}$$

Indeed, define

$$w = \sum_{i=1}^{r} \langle x, e_i \rangle \, e_i.$$

Then $w \in W$, and for every j,

$$\langle x - w, e_j \rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0.$$

Hence

$$x - w \in W^{\perp}.$$

Thus

$$w = P_W x.$$

If $W = \operatorname{span}(u)$ with $u \neq 0$, then

$$P_W x = \frac{\langle x, u \rangle}{\langle u, u \rangle} u.$$

2.14 Best Approximation Property

Orthogonal projection solves a natural minimization problem.

Theorem 2.14.1. Let $W \subseteq V$ be a subspace. For every $x \in V$,

$$P_W x$$

is the unique vector in W minimizing

$$\|x - w\|.$$

Proof. Let

$$p = P_W x.$$

For any $w \in W$,

$$x - w = (x - p) + (p - w).$$

Since

$$x - p \in W^{\perp}$$

and

$$p - w \in W,$$

the two terms are orthogonal. Therefore

$$\|x - w\|^2 = \|x - p\|^2 + \|p - w\|^2 .$$

Hence

$$\|x - w\| \geq \|x - p\| ,$$

with equality if and only if

$$p = w.$$

□

This result is the geometric foundation of the least-squares method.

2.15 Gram–Schmidt Orthogonalization

Every finite-dimensional inner product space admits an orthonormal basis.
The Gram–Schmidt process constructs one explicitly.
Suppose

$$v_1, \ldots, v_n$$

are linearly independent.
Define

$$u_1 = v_1.$$

Then remove from v_2 its component in the direction of u_1:

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1.$$

Next define

$$u_3 = v_3 - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2.$$

In general,

$$\boxed{u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j.}$$

Then

$$u_1, \ldots, u_n$$

are pairwise orthogonal.
Normalize them by setting

$$e_k = \frac{u_k}{\|u_k\|}.$$

Then

$$e_1, \ldots, e_n$$

form an orthonormal basis of

$$\operatorname{span}(v_1, \ldots, v_n).$$

Theorem 2.15.1 (Gram–Schmidt)**.** Let

$$v_1, \ldots, v_n$$

be linearly independent vectors in an inner product space. Then there exists an orthonormal collection

$$e_1, \ldots, e_n$$

such that

$$\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(v_1, \ldots, v_k)$$

for every $k = 1, \ldots, n$.

2.16 Example of Gram–Schmidt

Consider

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Take

$$u_1 = v_1.$$

Then

$$\|u_1\| = \sqrt{2},$$

so

$$e_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Next,

$$\langle v_2, e_1 \rangle = \frac{1}{\sqrt{2}}.$$

Hence

$$u_2 = v_2 - \langle v_2, e_1 \rangle \, e_1.$$

Therefore

$$u_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}.$$

Its norm is

$$\|u_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}.$$

Thus

$$e_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Therefore

$$e_1, e_2$$

form an orthonormal basis of

$$\operatorname{span}(v_1, v_2).$$

2.17 Existence of Orthonormal Bases

Corollary 2.17.1. Every finite-dimensional inner product space has an orthonormal basis.

Proof. Choose any basis

$$v_1, \dots, v_n$$

of V. Applying the Gram–Schmidt process produces an orthonormal basis

$$e_1, \dots, e_n.$$

□

More generally, an orthonormal set can be extended to an orthonormal basis.

Theorem 2.17.2. Let

$$e_1, \dots, e_r$$

be an orthonormal collection in V. Then it can be extended to an orthonormal basis of V.

Proof. Since the vectors are linearly independent, extend them to a basis

$$e_1, \dots, e_r, v_{r+1}, \dots, v_n.$$

Apply Gram–Schmidt while keeping the first r vectors unchanged. The resulting basis is orthonormal. □

2.18 Orthogonal and Unitary Matrices

Orthonormal bases can be encoded by matrices.
Let

$$Q = \begin{pmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{pmatrix} \in M_n(\mathbb{R}).$$

The columns of Q are orthonormal if and only if

$$Q^T Q = I.$$

Definition 2.18.1. A real square matrix Q is orthogonal if

$$Q^T Q = I.$$

Thus

$$Q^{-1} = Q^T.$$

For complex matrices, the corresponding notion is unitary.

Definition 2.18.2. A complex square matrix U is unitary if

$$U^* U = I.$$

Thus

$$U^{-1} = U^*.$$

The columns of a unitary matrix form an orthonormal basis of $\mathbb{C}^n$.
Similarly, the rows are also orthonormal.
Indeed, if

$$U^*U = I,$$

then U is invertible with

$$U^{-1} = U^*,$$

so

$$UU^* = I.$$

2.19 Orthogonal and Unitary Transformations

Let

$$T : V \to V$$

be a linear operator on an inner product space.

Definition 2.19.1. The operator T is an isometry if

$$\langle T(x), T(y)\rangle = \langle x, y\rangle$$

for all $x, y \in V$.

Thus an isometry preserves lengths:

$$\|T(x)\| = \|x\|,$$

and preserves orthogonality.
If $\mathcal{B}$ is an orthonormal basis and

$$A = [T]_{\mathcal{B}},$$

then T is an isometry if and only if

$$A^*A = I.$$

Therefore:

$$\boxed{\text{real isometries} \longleftrightarrow \text{orthogonal matrices},}$$

and

$$\boxed{\text{complex isometries} \longleftrightarrow \text{unitary matrices}.}$$

2.20 QR Decomposition

The Gram–Schmidt process has an important matrix interpretation.
Let

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix} \in M_{m\times n}(K),$$

and suppose the columns

$$a_1, \ldots, a_n$$

are linearly independent.

Apply Gram–Schmidt to obtain orthonormal vectors

$$q_1, \dots, q_n.$$

Each a_j belongs to

$$\operatorname{span}(q_1, \dots, q_j),$$

so

$$a_j = r_{1j}q_1 + \cdots + r_{jj}q_j.$$

Define

$$Q = \begin{pmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{pmatrix}$$

and

$$R = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & r_{nn} \end{pmatrix}.$$

Then

$$\boxed{A = QR.}$$

The columns of Q are orthonormal, so

$$Q^*Q = I.$$

Moreover,

$$Q^*A = Q^*QR = R.$$

Hence

$$\boxed{R = Q^*A.}$$

This is the QR decomposition produced by Gram–Schmidt.

Theorem 2.20.1 (QR Decomposition)**.** Let $A \in M_{m\times n}(K)$ have linearly independent columns. Then there exist a matrix $Q \in M_{m\times n}(K)$ with orthonormal columns and an invertible upper triangular matrix $R \in M_n(K)$ such that

$$A = QR.$$

If A is square and invertible, then Q is orthogonal over $\mathbb{R}$ or unitary over $\mathbb{C}$.

2.21 Projection Matrices

Let

$$Q \in M_{m\times r}(K)$$

have orthonormal columns.
Then the orthogonal projection onto the column space of Q is represented by

$$\boxed{P = QQ^*.}$$

Indeed, for any x,

$$Px = QQ^*x = \sum_{i=1}^{r} \langle x, q_i \rangle \, q_i.$$

The projection matrix satisfies

$$P^2 = P.$$

Also,

$$P^* = P.$$

Thus orthogonal projections are represented by self-adjoint idempotent matrices.
The significance of the condition

$$P^* = P$$

will become clearer after adjoint operators are introduced.

2.22 Least Squares and Orthogonal Projection

Consider an overdetermined system

$$Ax = b,$$

where

$$A \in M_{m\times n}(\mathbb{R}), \qquad m > n.$$

The system may have no exact solution.
Instead, we seek x minimizing

$$\|Ax - b\|.$$

Let

$$W = C(A)$$

be the column space of A.
Since

$$Ax \in W,$$

the problem is equivalent to finding the vector in W closest to b.
Therefore the optimal vector satisfies

$$Ax = P_W b.$$

The residual

$$r = b - Ax$$

must be orthogonal to the column space:

$$r \perp C(A).$$

Equivalently,

$$A^T(b - Ax) = 0.$$

Hence

$$\boxed{A^T Ax = A^T b.}$$

These are the normal equations.
If the columns of A are linearly independent, then

$$A^T A$$

is invertible, and the unique least-squares solution is

$$\boxed{x = (A^T A)^{-1} A^T b.}$$

2.23 Least Squares via QR Decomposition

Suppose

$$A = QR,$$

where the columns of Q are orthonormal and R is invertible upper triangular.
Then

$$Ax = QRx.$$

The least-squares problem

$$\min_x \|Ax - b\|$$

becomes

$$\min_x \|QRx - b\| .$$

The projection of b onto $C(A) = C(Q)$ is

$$QQ^*b.$$

Therefore

$$QRx = QQ^*b.$$

Multiplying by Q^*,

$$Rx = Q^*b.$$

Hence the least-squares solution satisfies

$$\boxed{Rx = Q^*b.}$$

Since R is upper triangular, this system can be solved efficiently by back substitution.
This illustrates how Gram–Schmidt, QR decomposition, orthogonal projection, and least squares are different aspects of the same geometric structure.

2.24 The Geometry Introduced by an Inner Product

The main ideas of this chapter form the chain

$$\boxed{\text{inner product} \longrightarrow \text{norm} \longrightarrow \text{orthogonality} \longrightarrow \text{orthonormal bases.}}$$

Orthogonal complements provide the decomposition

$$V = W \oplus W^{\perp},$$

and orthogonal projection selects the component of a vector lying in a chosen subspace.
The Gram–Schmidt process constructs orthonormal bases and, in matrix language, produces the QR decomposition

$$A = QR.$$

Least squares then arises naturally from projection:

$$\boxed{\text{best approximation} = \text{orthogonal projection.}}$$

The additional geometric structure of an inner product also allows us to associate a new operator to every linear map: its adjoint.
Adjoint operators will provide the language needed to define self-adjoint, unitary, and normal operators and will eventually lead to the spectral theorem.

Chapter 3

Adjoint and Normal Operators

3.1 The Adjoint of a Linear Map

Let V and W be finite-dimensional inner product spaces over $K = \mathbb{R}$ or $K = \mathbb{C}$.

Definition 3.1.1. Let

$$T : V \to W$$

be linear. The adjoint of T is the unique linear map

$$T^* : W \to V$$

such that

$$\langle T(v), w\rangle = \langle v, T^*(w)\rangle$$

for all $v \in V$ and $w \in W$.

Thus the adjoint transfers the action of T from the first slot of the inner product to the second slot.
The defining identity is

$$\boxed{\langle T(v), w\rangle = \langle v, T^*(w)\rangle\,.}$$

The existence and uniqueness of T^* rely on the finite-dimensional Riesz representation theorem.

3.2 Riesz Representation in Finite Dimensions

Let V be a finite-dimensional inner product space.

Theorem 3.2.1 (Riesz Representation Theorem)**.** For every linear functional

$$f : V \to K,$$

there exists a unique vector $y \in V$ such that

$$f(x) = \langle x, y\rangle$$

for every $x \in V$.

Proof. Let

$$(e_1, \ldots, e_n)$$

be an orthonormal basis of V.
For any

$$x = \sum_{i=1}^{n} x_i e_i,$$

linearity gives

$$f(x) = \sum_{i=1}^{n} x_i f(e_i).$$

Define

$$y = \sum_{i=1}^{n} \overline{f(e_i)}\, e_i.$$

Then

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i f(e_i) = f(x).$$

For uniqueness, suppose

$$\langle x, y\rangle = \langle x, z\rangle$$

for all $x \in V$. Then

$$\langle x, y - z\rangle = 0$$

for all $x \in V$. Taking $x = y - z$,

$$\|y - z\|^2 = 0,$$

so $y = z$. □

For fixed $w \in W$, the map

$$v \mapsto \langle T(v), w\rangle$$

is a linear functional on V. By Riesz representation, there exists a unique vector $T^*(w) \in V$ such that

$$\langle T(v), w\rangle = \langle v, T^*(w)\rangle .$$

This proves the existence and uniqueness of the adjoint.

3.3 Basic Properties of the Adjoint

Let

$$S, T : V \to W$$

be linear maps and let $a, b \in K$.
The adjoint satisfies

$$(aS + bT)^* = \overline{a}S^* + \overline{b}T^*.$$

Also,

$$(T^*)^* = T.$$

If

$$T : U \to V$$

and

$$S : V \to W,$$

then

$$(S \circ T)^* = T^* \circ S^*.$$

Thus the order reverses under taking adjoints.
For the identity map,

$$I^* = I.$$

If T is invertible, then

$$(T^{-1})^* = (T^*)^{-1}.$$

3.4 Matrix Representation of the Adjoint

Let

$$T : V \to W$$

be linear.
Choose orthonormal bases

$$\mathcal{B} = (v_1, \ldots, v_n)$$

of V and

$$\mathcal{C} = (w_1, \ldots, w_m)$$

of W.
Suppose

$$A = [T]_{\mathcal{C}\leftarrow\mathcal{B}}.$$

Theorem 3.4.1. With respect to the orthonormal bases $\mathcal{C}$ and $\mathcal{B}$,

$$\boxed{[T^*]_{\mathcal{B}\leftarrow\mathcal{C}} = A^*.}$$

Here

$$A^* = \overline{A}^T$$

is the conjugate transpose.
Over $\mathbb{R}$,

$$A^* = A^T.$$

Thus the abstract adjoint of a linear map corresponds exactly to the conjugate transpose of its matrix in orthonormal coordinates.

3.5 Kernel and Image Relations

The adjoint gives a precise relation between kernels, images, and orthogonal complements.

Theorem 3.5.1. Let

$$T : V \to W.$$

Then

$$\boxed{(\operatorname{Im} T)^\perp = \ker T^*.}$$

Proof. Let $w \in W$. Then

$$w \in (\operatorname{Im} T)^\perp$$

if and only if

$$\langle T(v), w\rangle = 0$$

for every $v \in V$.
By the defining property of the adjoint,

$$\langle v, T^*(w)\rangle = 0$$

for every $v \in V$. Hence

$$T^*(w) = 0.$$

Therefore

$$w \in \ker T^*.$$

□

Taking orthogonal complements gives

$$\boxed{\operatorname{Im} T = (\ker T^*)^\perp.}$$

Similarly,

$$\boxed{\operatorname{Im} T^* = (\ker T)^\perp.}$$

Therefore

$$V = \ker T \oplus \operatorname{Im} T^*,$$

and

$$W = \ker T^* \oplus \operatorname{Im} T.$$

These decompositions are fundamental in finite-dimensional spectral theory.

3.6 Self-Adjoint Operators

Let

$$T \in \mathrm{End}_K(V).$$

Definition 3.6.1. The operator T is self-adjoint if

$$T^* = T.$$

In an orthonormal basis, this condition becomes

$$A^* = A.$$

Thus over $\mathbb{R}$, self-adjoint operators are represented by symmetric matrices:

$$A^T = A.$$

Over $\mathbb{C}$, they are represented by Hermitian matrices:

$$A^* = A.$$

Proposition 3.6.2. If T is self-adjoint, then

$$\langle T(x), y\rangle = \langle x, T(y)\rangle$$

for all $x, y \in V$.

Theorem 3.6.3. Every eigenvalue of a self-adjoint operator is real.

Proof. Suppose

$$T(v) = \lambda v, \qquad v \neq 0.$$

Then

$$\langle T(v), v\rangle = \lambda \|v\|^2.$$

Since $T = T^*$,

$$\langle T(v), v\rangle = \langle v, T(v)\rangle = \overline{\lambda} \|v\|^2.$$

Therefore

$$\lambda = \overline{\lambda},$$

so $\lambda \in \mathbb{R}$. □

Theorem 3.6.4. Eigenvectors of a self-adjoint operator corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose

$$T(u) = \lambda u, \qquad T(v) = \mu v,$$

with $\lambda \neq \mu$.
Then

$$\langle T(u), v\rangle = \lambda \langle u, v\rangle.$$

Since T is self-adjoint,

$$\langle T(u), v\rangle = \langle u, T(v)\rangle = \mu \langle u, v\rangle.$$

Thus

$$(\lambda - \mu)\langle u, v\rangle = 0.$$

Since $\lambda \neq \mu$,

$$\langle u, v\rangle = 0.$$

□

3.7 Skew-Adjoint Operators

Definition 3.7.1. An operator T is skew-adjoint if

$$T^* = -T.$$

In a real orthonormal basis, this corresponds to

$$A^T = -A.$$

If

$$T(v) = \lambda v$$

for a nonzero complex vector v, then

$$\lambda = -\overline{\lambda}.$$

Hence every eigenvalue of a skew-adjoint operator is purely imaginary.
Over $\mathbb{R}$, nonzero skew-symmetric matrices therefore generally do not possess real eigenvalues.

3.8 Isometries

Definition 3.8.1. An operator

$$T \in \mathrm{End}_K(V)$$

is an isometry if

$$\|T(v)\| = \|v\|$$

for every $v \in V$.

Equivalently,

$$\langle T(u), T(v)\rangle = \langle u, v\rangle$$

for all $u, v \in V$.

Theorem 3.8.2. The operator T is an isometry if and only if

$$\boxed{T^*T = I.}$$

Since V is finite-dimensional, an isometry

$$T : V \to V$$

is automatically invertible.
Hence

$$T^{-1} = T^*,$$

and therefore also

$$TT^* = I.$$

In an orthonormal basis, an isometry is represented by a unitary matrix over $\mathbb{C}$ or an orthogonal matrix over $\mathbb{R}$.

3.9 Unitary and Orthogonal Operators

Definition 3.9.1. A complex linear operator U is unitary if

$$U^*U = UU^* = I.$$

Definition 3.9.2. A real linear operator Q is orthogonal if

$$Q^TQ = QQ^T = I.$$

Thus

$$U^{-1} = U^*$$

and

$$Q^{-1} = Q^T.$$

Unitary and orthogonal operators preserve inner products:

$$\langle U(x), U(y)\rangle = \langle x, y\rangle .$$

Consequently, they preserve:

$$\|x\| ,$$

angles, orthogonality, and distances.

Proposition 3.9.3. If U is unitary and

$$U(v) = \lambda v$$

for $v \neq 0$, then

$$|\lambda| = 1.$$

Proof. Since U preserves norms,

$$\|v\| = \|U(v)\| = \|\lambda v\| = |\lambda| \, \|v\| .$$

Since $v \neq 0$,

$$|\lambda| = 1.$$

□

Thus the eigenvalues of a unitary operator lie on the unit circle.

3.10 Positive Operators

Definition 3.10.1. A self-adjoint operator T is positive semidefinite if

$$\langle T(v), v\rangle \geq 0$$

for every $v \in V$.

We write

$$T \geq 0.$$

It is positive definite if

$$\langle T(v), v\rangle > 0$$

for every $v \neq 0$.
In an orthonormal basis, these notions correspond to Hermitian positive semidefinite and positive definite matrices.

Proposition 3.10.2. For every linear operator T,

$$T^*T$$

is positive semidefinite.

Indeed,

$$\langle T^*T(v), v\rangle = \langle T(v), T(v)\rangle = \|T(v)\|^2 \geq 0.$$

Similarly,

$$TT^*$$

is positive semidefinite.
These two operators will later play a central role in the singular value decomposition.

3.11 Normal Operators

Definition 3.11.1. An operator

$$T \in \mathrm{End}_K(V)$$

is normal if

$$\boxed{T^*T = TT^*.}$$

In an orthonormal basis, this condition becomes

$$A^*A = AA^*.$$

Important examples include:

- self-adjoint operators;
- skew-adjoint operators;
- unitary operators.

Indeed, if $T = T^*$, then

$$T^*T = T^2 = TT^*.$$

If $T^* = -T$, then

$$T^*T = -T^2 = TT^*.$$

If T is unitary, then

$$T^*T = TT^* = I.$$

Thus

$$\boxed{\text{self-adjoint,} \quad \text{skew-adjoint,} \quad \text{unitary} \quad \Longrightarrow \quad \text{normal.}}$$

3.12 Norm Equality for Normal Operators

Normality can be characterized by a useful norm identity.

Theorem 3.12.1. An operator T is normal if and only if

$$\|T(v)\| = \|T^*(v)\|$$

for every $v \in V$.

Proof. We have

$$\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle\,,$$

while

$$\|T^*(v)\|^2 = \langle T^*(v), T^*(v)\rangle = \langle TT^*(v), v\rangle\,.$$

Therefore

$$\|T(v)\|^2 - \|T^*(v)\|^2 = \langle (T^*T - TT^*)v, v\rangle\,.$$

If T is normal, the right-hand side is zero.
Conversely, if the norm equality holds for every v, then

$$\langle (T^*T - TT^*)v, v\rangle = 0$$

for every v. Since

$$T^*T - TT^*$$

is self-adjoint, this implies

$$T^*T - TT^* = 0.$$

Hence T is normal. □

3.13 Kernels of a Normal Operator and Its Adjoint

Corollary 3.13.1. If T is normal, then

$$\boxed{\ker T = \ker T^*.}$$

Indeed,

$$T(v) = 0 \iff \|T(v)\| = 0.$$

For normal T,

$$\|T(v)\| = \|T^*(v)\|\,,$$

so

$$T(v) = 0 \iff T^*(v) = 0.$$

Consequently,

$$\operatorname{Im} T = (\ker T^*)^\perp = (\ker T)^\perp.$$

Thus

$$\boxed{V = \ker T \oplus \operatorname{Im} T}$$

for a normal operator.

3.14 Eigenvectors of Normal Operators

Normal operators possess stronger eigenvector properties than general operators.

Proposition 3.14.1. Let T be normal. If

$$T(v) = \lambda v,$$

then

$$T^*(v) = \overline{\lambda} v.$$

Proof. Consider

$$S = T - \lambda I.$$

Since T is normal, S is also normal.
If

$$T(v) = \lambda v,$$

then

$$S(v) = 0.$$

For a normal operator,

$$\ker S = \ker S^*.$$

Thus

$$S^*(v) = 0.$$

But

$$S^* = T^* - \overline{\lambda} I.$$

Hence

$$T^*(v) = \overline{\lambda} v.$$

□

Theorem 3.14.2. Eigenvectors of a normal operator corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose

$$T(u) = \lambda u, \qquad T(v) = \mu v,$$

where

$$\lambda \neq \mu.$$

Then

$$T^*(v) = \overline{\mu} v.$$

Hence

$$\lambda \langle u, v \rangle = \langle T(u), v \rangle = \langle u, T^*(v) \rangle = \mu \langle u, v \rangle .$$

Therefore

$$(\lambda - \mu) \langle u, v \rangle = 0.$$

Since $\lambda \neq \mu$,

$$\langle u, v \rangle = 0.$$

□

3.15 Invariant Subspaces and Orthogonal Complements

For general operators, the orthogonal complement of an invariant subspace need not be invariant.
For normal operators, eigenspaces behave especially well.
Let

$$E_\lambda = \ker(T - \lambda I).$$

Then

$$E_\lambda^\perp$$

is T-invariant.
Indeed, if

$$x \in E_\lambda^\perp$$

and

$$v \in E_\lambda,$$

then

$$T^*(v) = \overline{\lambda} v.$$

Therefore

$$\langle T(x), v \rangle = \langle x, T^*(v) \rangle = \left\langle x, \overline{\lambda} v \right\rangle = \lambda \langle x, v \rangle = 0.$$

Hence

$$T(x) \in E_\lambda^\perp.$$

This property makes it possible to construct orthonormal eigenbases inductively.

3.16 Normal Matrices

Let

$$A \in M_n(\mathbb{C}).$$

Definition 3.16.1. The matrix A is normal if

$$A^* A = A A^*.$$

Special cases include:

$$A = A^*$$

for Hermitian matrices,

$$A^* = -A$$

for skew-Hermitian matrices, and

$$A^* A = I$$

for unitary matrices.
Over $\mathbb{R}$, symmetric and orthogonal matrices are normal.

Example 3.16.2. The matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

is real orthogonal and hence normal.
It is not symmetric, since

$$A^T = -A.$$

Thus normality is a broader condition than self-adjointness.

3.17 Unitary Similarity

In ordinary linear algebra, changing an arbitrary basis transforms a matrix by

$$A \longmapsto P^{-1}AP.$$

In an inner product space, if both bases are orthonormal, the change-of-basis matrix U is unitary. Hence

$$U^{-1} = U^*.$$

Therefore the transformation becomes

$$\boxed{A \longmapsto U^*AU.}$$

Definition 3.17.1. Matrices A and B are unitarily similar if

$$B = U^*AU$$

for some unitary matrix U.

Over $\mathbb{R}$, the analogous notion is orthogonal similarity:

$$B = Q^TAQ.$$

Unitary similarity is more restrictive than ordinary similarity because it preserves the inner product geometry.

3.18 Schur Triangularization

Before proving the spectral theorem, it is useful to know that every complex matrix can be triangularized using a unitary change of basis.

Theorem 3.18.1 (Schur Triangularization)**.** Let

$$A \in M_n(\mathbb{C}).$$

Then there exists a unitary matrix U such that

$$\boxed{U^*AU = R,}$$

where R is upper triangular.

The diagonal entries of R are the eigenvalues of A, counted with algebraic multiplicity.

Proof. We proceed by induction on n.
Since the characteristic polynomial splits over $\mathbb{C}$, A has an eigenvalue λ and a unit eigenvector u_1.
Extend u_1 to an orthonormal basis

$$u_1, \ldots, u_n.$$

Because

$$A(u_1) = \lambda u_1,$$

the matrix of A in this basis has the form

$$\begin{pmatrix} \lambda & * \\ 0 & B \end{pmatrix}.$$

Apply the induction hypothesis to B. The resulting orthonormal change of basis on the remaining coordinates produces an upper triangular matrix for A. □

Thus every complex matrix is unitarily similar to an upper triangular matrix.

3.19 Normality and Triangular Matrices

The key observation behind the spectral theorem is that a triangular normal matrix must already be diagonal.

Lemma 3.19.1. If an upper triangular matrix R is normal, then R is diagonal.

Proof. Write

$$R = (r_{ij}).$$

Since R is normal,

$$R^*R = RR^*.$$

Compare the $(1,1)$-entries:

$$\sum_{k=1}^{n} |r_{k1}|^2 = \sum_{k=1}^{n} |r_{1k}|^2 .$$

Since R is upper triangular,

$$r_{k1} = 0$$

for $k > 1$. Therefore

$$|r_{11}|^2 = |r_{11}|^2 + \sum_{k=2}^{n} |r_{1k}|^2 .$$

Hence

$$r_{1k} = 0$$

for every $k > 1$.
Repeating the same argument for the lower-right principal submatrix shows that every off-diagonal entry vanishes. □

This lemma connects normality with unitary diagonalization.

3.20 Toward the Spectral Theorem

We now have the ingredients needed for the central result of spectral theory.
Every complex matrix admits a Schur decomposition

$$A = URU^*,$$

where R is upper triangular.
If A is normal, then R is also normal because

$$R = U^*AU.$$

But an upper triangular normal matrix must be diagonal.
Therefore a normal matrix is unitarily diagonalizable.
Conversely, every matrix of the form

$$A = UDU^*$$

with D diagonal is normal.
This gives the fundamental equivalence

$$\boxed{A \text{ is normal } \iff A \text{ is unitarily diagonalizable.}}$$

The full spectral theorem, together with its important special cases for Hermitian, real symmetric, and unitary matrices, will be developed in the next chapter.

Chapter 4

Spectral Theorems

4.1 The Spectral Problem

Let V be a finite-dimensional inner product space over $K = \mathbb{R}$ or $K = \mathbb{C}$, and let

$$T \in \mathrm{End}_K(V).$$

The ordinary diagonalization problem asks whether there exists a basis of eigenvectors of T.
In an inner product space, a stronger question is available:

$$\boxed{\text{Does } T \text{ admit an orthonormal basis of eigenvectors?}}$$

If the answer is yes, then relative to an orthonormal basis,

$$[T] = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Equivalently, for a complex matrix A, we seek a unitary matrix U such that

$$U^* A U = D,$$

where D is diagonal.
Over $\mathbb{R}$, we seek an orthogonal matrix Q such that

$$Q^T A Q = D.$$

Because unitary and orthogonal transformations preserve the inner product, this form of diagonalization preserves the geometry of the space.

4.2 The Spectral Theorem for Normal Operators

Recall that a complex linear operator T is normal if

$$T^* T = T T^*.$$

The central result is the following.

Theorem 4.2.1 (Spectral Theorem for Normal Operators)**.** Let V be a finite-dimensional complex inner product space, and let

$$T \in \mathrm{End}_{\mathbb{C}}(V).$$

Then the following are equivalent:

1. T is normal;
2. V has an orthonormal basis consisting of eigenvectors of T;
3. T is unitarily diagonalizable.

Equivalently, for $A \in M_n(\mathbb{C})$,

$$\boxed{A^* A = A A^* \iff A = U D U^*}$$

for some unitary U and diagonal D.

Proof. Suppose first that T is normal.
By Schur triangularization, there exists an orthonormal basis in which the matrix of T is upper triangular:

$$[T] = R.$$

Since unitary similarity preserves normality, R is normal.
From the previous chapter, an upper triangular normal matrix must be diagonal. Hence

$$R = D$$

for some diagonal matrix D.
Thus T has an orthonormal eigenbasis.
Conversely, suppose

$$T = UDU^*$$

with U unitary and D diagonal. Then

$$T^* = UD^*U^*.$$

Therefore

$$TT^* = UDD^*U^*$$

and

$$T^*T = UD^*DU^*.$$

Since diagonal matrices commute with their adjoints,

$$DD^* = D^*D.$$

Hence

$$TT^* = T^*T.$$

Therefore T is normal. □

This theorem gives a complete characterization of those operators that can be diagonalized without destroying the inner product structure.

4.3 Orthogonal Decomposition into Eigenspaces

Let T be normal, with distinct eigenvalues

$$\lambda_1, \ldots, \lambda_r.$$

From the previous chapter, eigenspaces corresponding to distinct eigenvalues are orthogonal.
Therefore

$$\boxed{V = E_{\lambda_1} \oplus^{\perp} \cdots \oplus^{\perp} E_{\lambda_r},}$$

where $\oplus^{\perp}$ denotes an orthogonal direct sum.
Choose an orthonormal basis of each eigenspace. Combining these bases gives an orthonormal eigenbasis of V.
Thus the spectral theorem may also be understood as an orthogonal decomposition of the entire space into eigenspaces.

4.4 Spectral Projections

Let

$$V = E_{\lambda_1} \oplus^{\perp} \cdots \oplus^{\perp} E_{\lambda_r}$$

be the eigenspace decomposition of a normal operator.
Let

$$P_i : V \to E_{\lambda_i}$$

denote the orthogonal projection onto E_{λ_i}.
Then

$$P_i^* = P_i, \qquad P_i^2 = P_i.$$

Moreover,

$$P_i P_j = 0$$

for $i \neq j$, and

$$P_1 + \cdots + P_r = I.$$

Every vector decomposes uniquely as

$$v = P_1 v + \cdots + P_r v.$$

Since T acts as multiplication by λ_i on E_{λ_i},

$$T(v) = \sum_{i=1}^{r} \lambda_i P_i v.$$

Therefore

$$\boxed{T = \sum_{i=1}^{r} \lambda_i P_i.}$$

This is called the spectral decomposition of T.

4.5 Polynomial Functions of a Normal Operator

If

$$T = \sum_{i=1}^{r} \lambda_i P_i,$$

then

$$T^k = \sum_{i=1}^{r} \lambda_i^k P_i.$$

Hence for any polynomial p,

$$\boxed{p(T) = \sum_{i=1}^{r} p(\lambda_i) P_i.}$$

In matrix form, if

$$A = UDU^*,$$

then

$$p(A) = U p(D) U^*,$$

where

$$p(D) = \operatorname{diag}\Big(p(\lambda_1), \ldots, p(\lambda_n)\Big).$$

Thus polynomial calculations involving a normal matrix reduce to scalar calculations involving its eigenvalues.

4.6 Self-Adjoint Operators

A self-adjoint operator satisfies

$$T^* = T.$$

Every self-adjoint operator is normal, so the spectral theorem applies.
In addition, all eigenvalues are real.
Therefore:

Theorem 4.6.1 (Spectral Theorem for Self-Adjoint Operators)**.** Let V be a finite-dimensional complex inner product space and let $T \in \mathrm{End}_{\mathbb{C}}(V)$ be self-adjoint. Then:

1. every eigenvalue of T is real;
2. eigenspaces associated with distinct eigenvalues are orthogonal;
3. V has an orthonormal basis consisting of eigenvectors of T;
4. there exists a unitary matrix U such that

$$U^*[T]U = \mathrm{diag}(\lambda_1, \ldots, \lambda_n),$$

   where every $\lambda_i \in \mathbb{R}$.

Thus a Hermitian matrix has a unitary diagonalization with real diagonal entries.

4.7 Real Symmetric Matrices

The real version is especially important.

Theorem 4.7.1 (Real Spectral Theorem)**.** Let

$$A \in M_n(\mathbb{R})$$

be symmetric:

$$A^T = A.$$

Then there exists an orthogonal matrix Q and a real diagonal matrix D such that

$$\boxed{Q^T A Q = D.}$$

Equivalently,

$$A = QDQ^T.$$

The columns of Q form an orthonormal basis of real eigenvectors of A.
Thus every real symmetric matrix is diagonalizable over $\mathbb{R}$, even though a general real matrix may fail to have any real eigenvalues.

Remark 4.7.2. This is stronger than ordinary diagonalization. Ordinary diagonalization only requires

$$A = PDP^{-1}$$

for some invertible P. The real spectral theorem allows P to be chosen orthogonal:

$$P^{-1} = P^T.$$

4.8 Hermitian Matrices

A complex matrix A is Hermitian if

$$A^* = A.$$

By the spectral theorem,

$$A = UDU^*,$$

where U is unitary and

$$D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$$

has real diagonal entries.
Hence

$$\boxed{A \text{ Hermitian} \Longrightarrow \operatorname{Spec}(A) \subseteq \mathbb{R}.}$$

Moreover, eigenvectors associated with different eigenvalues may be chosen orthogonal.

4.9 Skew-Hermitian Matrices

Suppose

$$A^* = -A.$$

Then

$$iA$$

is Hermitian because

$$(iA)^* = -iA^* = iA.$$

Therefore iA is unitarily diagonalizable with real eigenvalues.
It follows that every eigenvalue of A is purely imaginary.
Thus there exists a unitary matrix U such that

$$U^*AU = \operatorname{diag}(i\mu_1, \dots, i\mu_n),$$

where

$$\mu_i \in \mathbb{R}.$$

4.10 Unitary Operators

A unitary operator satisfies

$$T^*T = TT^* = I.$$

Hence every unitary operator is normal and therefore unitarily diagonalizable.

Theorem 4.10.1. Let U be a unitary operator on a finite-dimensional complex inner product space. Then there exists an orthonormal basis such that

$$[U] = \operatorname{diag}(\lambda_1, \dots, \lambda_n),$$

where

$$|\lambda_i| = 1.$$

Thus the spectrum of a unitary operator lies on the unit circle:

$$\boxed{\operatorname{Spec}(U) \subseteq \{z \in \mathbb{C} : |z| = 1\}.}$$

Every eigenvalue may therefore be written as

$$\lambda_j = e^{i\theta_j}.$$

4.11 Orthogonal Matrices

A real orthogonal matrix satisfies

$$Q^T Q = I.$$

When regarded as a complex matrix, Q is unitary. Thus its complex eigenvalues have modulus 1.
However, a real orthogonal matrix need not be diagonalizable over $\mathbb{R}$.
For example,

$$Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

represents a rotation by 90°. Its eigenvalues are

$$i \qquad \text{and} \qquad -i.$$

Thus Q is diagonalizable over $\mathbb{C}$, but not over $\mathbb{R}$.
This distinction reflects the fact that the complex spectral theorem for normal matrices does not automatically imply real diagonalizability.

4.12 Positive Semidefinite Operators

Let T be self-adjoint.
Recall that T is positive semidefinite if

$$\langle T(v), v \rangle \geq 0$$

for every $v \in V$.
The spectral theorem gives a simple eigenvalue characterization.

Theorem 4.12.1. Let T be self-adjoint. Then

$$T \geq 0$$

if and only if every eigenvalue of T is nonnegative.

Proof. Choose an orthonormal eigenbasis

$$e_1, \ldots, e_n$$

with

$$T(e_i) = \lambda_i e_i.$$

For

$$v = \sum_{i=1}^{n} a_i e_i,$$

we obtain

$$\langle T(v), v \rangle = \sum_{i=1}^{n} \lambda_i |a_i|^2 .$$

If every $\lambda_i \geq 0$, this quantity is nonnegative.
Conversely, taking $v = e_i$ gives

$$\lambda_i = \langle T(e_i), e_i \rangle \geq 0.$$

□

Similarly,

$$T > 0$$

if and only if every eigenvalue is strictly positive.

4.13 Square Roots of Positive Operators

Let $T \geq 0$.
By the spectral theorem,

$$T = UDU^*,$$

where

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_i \geq 0.$$

Define

$$D^{1/2} = \operatorname{diag}\Big(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}\Big).$$

Then

$$T^{1/2} = UD^{1/2}U^*.$$

It satisfies

$$(T^{1/2})^2 = T$$

and

$$T^{1/2} \geq 0.$$

Theorem 4.13.1. Every positive semidefinite operator has a unique positive semidefinite square root.

This construction will later be useful in the polar decomposition and the singular value decomposition.

4.14 Rayleigh Quotient

Let $A \in M_n(\mathbb{R})$ be symmetric, or more generally let A be Hermitian.
For $x \neq 0$, define the Rayleigh quotient by

$$\boxed{R_A(x) = \frac{x^*Ax}{x^*x}.}$$

If

$$\|x\| = 1,$$

then

$$R_A(x) = x^*Ax.$$

Suppose

$$A = UDU^*,$$

where

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

and

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Write

$$x = \sum_{i=1}^{n} c_i u_i,$$

where u_i are orthonormal eigenvectors.
Then

$$x^*Ax = \sum_{i=1}^{n} \lambda_i \, |c_i|^2$$

and

$$x^*x = \sum_{i=1}^{n} |c_i|^2 .$$

Therefore

$$R_A(x) = \frac{\sum_{i=1}^{n} \lambda_i |c_i|^2}{\sum_{i=1}^{n} |c_i|^2}.$$

Thus the Rayleigh quotient is a weighted average of the eigenvalues.
Consequently,

$$\boxed{\lambda_n \leq R_A(x) \leq \lambda_1.}$$

4.15 Extremal Characterization of Eigenvalues

The preceding observation gives the first variational principle.

Theorem 4.15.1. Let A be Hermitian, with eigenvalues

$$\lambda_1 \geq \cdots \geq \lambda_n.$$

Then

$$\boxed{\lambda_1 = \max_{\|x\|=1} x^*Ax}$$

and

$$\boxed{\lambda_n = \min_{\|x\|=1} x^*Ax.}$$

The maximum is attained precisely on the eigenspace corresponding to λ_1, while the minimum is attained on the eigenspace corresponding to λ_n.
Thus eigenvalues may be characterized without solving the characteristic equation directly.

4.16 Restriction to Orthogonal Complements

Let

$$u_1, \ldots, u_n$$

be an orthonormal eigenbasis satisfying

$$Au_i = \lambda_i u_i,$$

with

$$\lambda_1 \geq \cdots \geq \lambda_n.$$

If

$$x \perp u_1,$$

then

$$x = \sum_{i=2}^{n} c_i u_i.$$

Hence

$$R_A(x) \leq \lambda_2.$$

Therefore

$$\lambda_2 = \max_{\substack{\|x\|=1 \\ x \perp u_1}} x^*Ax.$$

More generally,

$$\boxed{\lambda_k = \max_{\substack{\|x\|=1 \\ x \perp u_1, \ldots, u_{k-1}}} x^* A x.}$$

This gives a recursive variational construction of all eigenvalues.

4.17 Courant–Fischer Min–Max Principle

The preceding characterization can be expressed without explicitly referring to eigenvectors.
Let

$$A = A^*$$

have eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Theorem 4.17.1 (Courant–Fischer)**.** For $k = 1, \ldots, n$,

$$\boxed{\lambda_k = \max_{\substack{S \subseteq K^n \\ \dim S = k}} \min_{\substack{x \in S \\ \|x\|=1}} x^* A x.}$$

Equivalently,

$$\boxed{\lambda_k = \min_{\substack{S \subseteq K^n \\ \dim S = n-k+1}} \max_{\substack{x \in S \\ \|x\|=1}} x^* A x.}$$

The first expression is called a max–min characterization, while the second is a min–max characterization.
For $k = 1$, the first formula becomes

$$\lambda_1 = \max_{\|x\|=1} x^* A x.$$

For $k = n$, the second becomes

$$\lambda_n = \min_{\|x\|=1} x^* A x.$$

The theorem shows that the entire ordered spectrum is encoded in the geometry of the quadratic quantity

$$x^* A x.$$

4.18 Spectral Norm of a Hermitian Matrix

For a Hermitian matrix A, the operator norm induced by the Euclidean norm is

$$\|A\| = \max_{\|x\|=1} \|Ax\|.$$

Using the spectral decomposition,

$$A = UDU^*,$$

we obtain

$$\boxed{\|A\| = \max_i |\lambda_i|.}$$

Thus the operator norm of a Hermitian matrix is the largest absolute value of its eigenvalues.
If $A \geq 0$, then every eigenvalue is nonnegative, so

$$\|A\| = \lambda_{\max}(A).$$

4.19 Spectral Decomposition of a Matrix

Let

$$A = A^*$$

and let

$$u_1, \ldots, u_n$$

be an orthonormal eigenbasis with

$$Au_i = \lambda_i u_i.$$

Define

$$P_i = u_i u_i^*.$$

Then P_i is the orthogonal projection onto

$$\operatorname{span}(u_i).$$

Therefore

$$\boxed{A = \sum_{i=1}^{n} \lambda_i u_i u_i^*.}$$

If eigenvalues are repeated, it is often more natural to group together equal eigenvalues:

$$A = \sum_{\lambda \in \operatorname{Spec}(A)} \lambda P_\lambda,$$

where P_λ is the orthogonal projection onto the eigenspace E_λ.
This expression makes the geometry of the spectral theorem explicit.

4.20 Spectral Theorem versus Jordan Theory

Jordan normal form and the spectral theorem both simplify linear operators, but they solve different problems.
Jordan normal form uses arbitrary invertible changes of basis:

$$A \longmapsto P^{-1}AP.$$

It applies to every matrix whose characteristic polynomial splits, even when the matrix is not diagonalizable.
The spectral theorem uses unitary or orthogonal changes of basis:

$$A \longmapsto U^*AU.$$

It applies only to special classes of operators, most generally normal operators over $\mathbb{C}$, but gives the stronger conclusion of diagonal form with an orthonormal eigenbasis.
Thus

$$\boxed{\text{Jordan theory} = \text{general similarity classification,}}$$

while

$$\boxed{\text{spectral theory} = \text{orthogonal/unitary diagonalization.}}$$

For a normal operator, every Jordan block has size 1. Hence its Jordan normal form is already diagonal.

4.21 Summary of the Spectral Theorem

The main equivalence of this chapter is

$$\boxed{T \text{ normal} \iff T \text{ has an orthonormal eigenbasis} \iff T \text{ is unitarily diagonalizable.}}$$

Important special cases include:

$$\boxed{T = T^* \Longrightarrow \text{real spectrum},}$$

$$\boxed{A = A^T \Longrightarrow A = QDQ^T,}$$

and

$$\boxed{U^*U = I \Longrightarrow |\lambda| = 1 \text{ for every eigenvalue } \lambda.}$$

For Hermitian matrices, the spectral theorem also gives the variational interpretation

$$\lambda_{\max} = \max_{\|x\|=1} x^*Ax,$$

$$\lambda_{\min} = \min_{\|x\|=1} x^*Ax,$$

with the remaining eigenvalues described by max–min and min–max principles.
Thus the spectrum is not merely the collection of roots of a characteristic polynomial. For self-adjoint operators, it describes the geometry and extremal behavior of the quadratic quantity

$$x^*Ax.$$

The next part of the volume turns from linear operators to a different kind of linear-algebraic object: bilinear and quadratic forms. Their natural transformation law is not similarity but congruence, leading to a new classification problem.

Chapter 5

Bilinear Forms

5.1 Bilinear Forms

Let V and W be finite-dimensional vector spaces over a field K.

Definition 5.1.1. A bilinear form on $V \times W$ is a map

$$B : V \times W \to K$$

that is linear in each variable separately.

Thus for all $x_1, x_2 \in V$, $y_1, y_2 \in W$, and $a, b \in K$,

$$B(ax_1 + bx_2, y) = aB(x_1, y) + bB(x_2, y),$$

and

$$B(x, ay_1 + by_2) = aB(x, y_1) + bB(x, y_2).$$

When $V = W$, we simply call

$$B : V \times V \to K$$

a bilinear form on V.

Example 5.1.2. On K^n, let

$$A \in M_n(K).$$

Then

$$B(x, y) = x^T A y$$

defines a bilinear form.

Example 5.1.3. On $M_n(K)$, the map

$$B(A, B) = \operatorname{tr}(AB)$$

is bilinear.

Example 5.1.4. Let V be a space of real-valued continuous functions on $[a, b]$. Then

$$B(f, g) = \int_a^b f(x)g(x)\,dx$$

is bilinear.

5.2 Matrix Representation

Let

$$\mathcal{B} = (v_1, \ldots, v_m)$$

be a basis of V, and let

$$\mathcal{C} = (w_1, \ldots, w_n)$$

be a basis of W.

Definition 5.2.1. The matrix of a bilinear form

$$B : V \times W \to K$$

with respect to $\mathcal{B}$ and $\mathcal{C}$ is

$$G = \Big(B(v_i, w_j)\Big) \in M_{m \times n}(K).$$

Let

$$x = \sum_{i=1}^{m} x_i v_i, \qquad y = \sum_{j=1}^{n} y_j w_j.$$

Then

$$B(x, y) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j B(v_i, w_j).$$

Therefore

$$\boxed{B(x, y) = [x]_{\mathcal{B}}^T G [y]_{\mathcal{C}}.}$$

Thus every finite-dimensional bilinear form is represented by a matrix.
Conversely, every matrix $G \in M_{m \times n}(K)$ defines a bilinear form through

$$B(x, y) = x^T G y$$

after bases have been fixed.

5.3 Gram Matrices of Bilinear Forms

When

$$B : V \times V \to K$$

and the same basis

$$\mathcal{B} = (v_1, \ldots, v_n)$$

is used in both variables, the representing matrix

$$G_{\mathcal{B}}(B) = \Big(B(v_i, v_j)\Big)$$

is called the Gram matrix of B relative to $\mathcal{B}$.
Then

$$B(x, y) = [x]_{\mathcal{B}}^T G_{\mathcal{B}}(B) [y]_{\mathcal{B}}.$$

Remark 5.3.1. The Gram matrix depends on the chosen basis. The bilinear form itself does not.

For the standard dot product on $\mathbb{R}^n$ with the standard basis,

$$G = I_n.$$

5.4 Change of Basis and Congruence

Let

$$B : V \times V \to K$$

be bilinear.
Suppose $\mathcal{B}$ and $\mathcal{C}$ are two bases of V, and let P be the change-of-basis matrix satisfying

$$[x]_{\mathcal{B}} = P [x]_{\mathcal{C}}.$$

Let

$$G_{\mathcal{B}}$$

and

$$G_{\mathcal{C}}$$

be the Gram matrices of B in the two bases.
Then

$$B(x,y) = [x]_{\mathcal{B}}^T G_{\mathcal{B}} [y]_{\mathcal{B}}.$$

Substituting

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{C}}$$

and

$$[y]_{\mathcal{B}} = P[y]_{\mathcal{C}},$$

we obtain

$$B(x,y) = [x]_{\mathcal{C}}^T P^T G_{\mathcal{B}} P [y]_{\mathcal{C}}.$$

Hence

$$\boxed{G_{\mathcal{C}} = P^T G_{\mathcal{B}} P.}$$

Definition 5.4.1. Two matrices $A, B \in M_n(K)$ are congruent if there exists an invertible matrix P such that

$$B = P^T A P.$$

Congruence is therefore the natural equivalence relation for matrices representing bilinear forms.
This should be contrasted with similarity:

$$A \longmapsto P^{-1} A P.$$

Similarity corresponds to changing coordinates for a linear operator, while congruence corresponds to changing coordinates in both arguments of a bilinear form.

5.5 Similarity versus Congruence

The two transformation laws

$$P^{-1} A P$$

and

$$P^T A P$$

must be distinguished carefully.
For a linear operator

$$T : V \to V,$$

a change of basis gives

$$[T]_{\mathcal{C}} = P^{-1} [T]_{\mathcal{B}} P.$$

For a bilinear form

$$B : V \times V \to K,$$

the same change of basis gives

$$G_{\mathcal{C}} = P^T G_{\mathcal{B}} P.$$

Thus

$$\boxed{\text{linear operators} \longleftrightarrow \text{similarity},}$$

while

$$\boxed{\text{bilinear forms} \longleftrightarrow \text{congruence}.}$$

The corresponding classification problems are therefore different.

5.6 Transpose of a Bilinear Form

Let

$$B : V \times V \to K$$

be bilinear.

Definition 5.6.1. The transpose of B is the bilinear form

$$B^T : V \times V \to K$$

defined by

$$B^T(x, y) = B(y, x).$$

If G is the Gram matrix of B, then

$$G^T$$

is the Gram matrix of B^T.
Thus properties of B involving interchange of its arguments are reflected by corresponding matrix symmetries.

5.7 Symmetric Bilinear Forms

Definition 5.7.1. A bilinear form B is symmetric if

$$B(x, y) = B(y, x)$$

for all $x, y \in V$.

Equivalently,

$$B^T = B.$$

If G is the Gram matrix of B, then

$$\boxed{B \text{ symmetric } \iff G^T = G.}$$

Thus symmetric bilinear forms correspond exactly to symmetric matrices.

Example 5.7.2. For a symmetric matrix

$$A = A^T,$$

the form

$$B(x, y) = x^T A y$$

is symmetric.

5.8 Alternating Bilinear Forms

Definition 5.8.1. A bilinear form B is alternating if

$$B(x, x) = 0$$

for every $x \in V$.

If the characteristic of K is not 2, then every alternating form is skew-symmetric. Indeed,

$$0 = B(x + y, x + y).$$

Expanding gives

$$0 = B(x, x) + B(x, y) + B(y, x) + B(y, y).$$

Since

$$B(x, x) = B(y, y) = 0,$$

we obtain

$$B(x, y) = -B(y, x).$$

Hence

$$B^T = -B.$$

In matrix form,

$$G^T = -G.$$

In particular, all diagonal entries vanish:

$$g_{ii} = 0.$$

Remark 5.8.2. Alternating bilinear forms belong to standard linear algebra. More advanced geometric structures built from nondegenerate alternating forms are not developed in this volume.

5.9 Symmetric and Alternating Parts

Assume

$$\operatorname{char} K \neq 2.$$

Every bilinear form B can be decomposed uniquely as

$$B = B_s + B_a,$$

where

$$B_s(x, y) = \frac{1}{2}\Big(B(x, y) + B(y, x)\Big)$$

is symmetric, and

$$B_a(x, y) = \frac{1}{2}\Big(B(x, y) - B(y, x)\Big)$$

is alternating.
Thus

$$\boxed{B = \frac{B + B^T}{2} + \frac{B - B^T}{2}.}$$

At the matrix level,

$$G = \frac{G + G^T}{2} + \frac{G - G^T}{2}.$$

The first matrix is symmetric and the second is skew-symmetric.

5.10 Left and Right Radicals

For a general bilinear form

$$B : V \times W \to K,$$

degeneracy can occur in either variable.

Definition 5.10.1. The left radical of B is

$$\operatorname{rad}_L(B) = \{x \in V : B(x, y) = 0 \text{ for all } y \in W\}.$$

Definition 5.10.2. The right radical of B is

$$\operatorname{rad}_R(B) = \{y \in W : B(x, y) = 0 \text{ for all } x \in V\}.$$

Both are subspaces.
Suppose the matrix of B is

$$G \in M_{m\times n}(K).$$

Then

$$x \in \operatorname{rad}_L(B)$$

if and only if

$$x^T G = 0,$$

while

$$y \in \operatorname{rad}_R(B)$$

if and only if

$$Gy = 0.$$

Therefore the radicals correspond to the left and right null spaces of the representing matrix.
This is the matrix interpretation recorded in the original notes: the left radical corresponds to the left null space of the Gram matrix, while the right radical corresponds to its null space. :contentReference[oaicite:2]index=2

5.11 Degenerate and Nondegenerate Forms

Definition 5.11.1. A bilinear form

$$B : V \times W \to K$$

is left nondegenerate if

$$\operatorname{rad}_L(B) = \{0\}.$$

It is right nondegenerate if

$$\operatorname{rad}_R(B) = \{0\}.$$

When both radicals vanish, B is called nondegenerate.
If

$$V = W$$

is finite-dimensional and the same basis is used in both variables, then the representing matrix is square.

Theorem 5.11.2. Let V be finite-dimensional, and let B be a bilinear form on V with Gram matrix G. Then the following are equivalent:

1. B is nondegenerate;
2. $$\operatorname{rad}_L(B) = \{0\};$$
3. $$\operatorname{rad}_R(B) = \{0\};$$
4. G is invertible;
5. $$\det G \neq 0.$$

The original notes explicitly identify this equivalence between nondegeneracy and invertibility of the Gram matrix. :contentReference[oaicite:3]index=3

5.12 Orthogonality for Bilinear Forms

Let B be a bilinear form on V.
We say that x is left orthogonal to y if

$$B(x, y) = 0.$$

For symmetric bilinear forms, left and right orthogonality coincide, so we simply write

$$x \perp_B y.$$

For a subspace $U \subseteq V$, define

$$U^{\perp} = \{v \in V : B(v, u) = 0 \text{ for all } u \in U\}$$

when B is symmetric.
Then

$$U^{\perp}$$

is a subspace of V.

Proposition 5.12.1. If B is nondegenerate and symmetric, then

$$\dim U + \dim U^{\perp} = \dim V.$$

This resembles the dimension formula for orthogonal complements in an inner product space, but positive definiteness is not required here.

5.13 Nondegenerate Subspaces

Let B be a symmetric bilinear form on V.
A subspace $U \subseteq V$ is called nondegenerate if the restriction

$$B|_{U \times U}$$

is nondegenerate.

Theorem 5.13.1. A subspace U is nondegenerate if and only if

$$U \cap U^{\perp} = \{0\}.$$

If B itself is nondegenerate and U is nondegenerate, then

$$\boxed{V = U \oplus U^{\perp}.}$$

Thus nondegenerate subspaces admit orthogonal complements.
This gives a useful method for decomposing bilinear spaces into smaller pieces.

5.14 Orthogonal Direct Sums

Let $U, W \subseteq V$.
We write

$$V = U \perp W$$

if

$$V = U \oplus W$$

and

$$B(u, w) = 0$$

for every $u \in U$ and $w \in W$.
More generally,

$$V = V_1 \perp \cdots \perp V_r$$

means that

$$V = V_1 \oplus \cdots \oplus V_r$$

and the subspaces are pairwise orthogonal with respect to B.
If bases are chosen separately in each V_i, the Gram matrix has block diagonal form:

$$G = \begin{pmatrix} G_1 & & 0 \\ & \ddots & \\ 0 & & G_r \end{pmatrix}.$$

Thus orthogonal decomposition of a bilinear space corresponds to block diagonalization of its Gram matrix under congruence.

5.15 Diagonalization of Symmetric Bilinear Forms

Assume

$$\operatorname{char} K \neq 2.$$

Let B be a symmetric bilinear form on a finite-dimensional vector space V.
A fundamental result states that B can be diagonalized by a suitable choice of basis.

Theorem 5.15.1. There exists a basis

$$(v_1, \ldots, v_n)$$

of V such that

$$B(v_i, v_j) = 0$$

whenever $i \neq j$.

Equivalently, every symmetric matrix over such a field is congruent to a diagonal matrix. Thus for every symmetric

$$A \in M_n(K),$$

there exists an invertible matrix P such that

$$\boxed{P^T A P = D,}$$

where D is diagonal.
This is not the spectral theorem. The matrix P is only required to be invertible, not orthogonal or unitary.

5.16 Constructing an Orthogonal Basis

Suppose there exists

$$v_1 \in V$$

such that

$$B(v_1, v_1) \neq 0.$$

Then

$$U_1 = \operatorname{span}(v_1)$$

is nondegenerate.
Hence

$$V = U_1 \oplus U_1^{\perp}.$$

Restrict B to $U_1^{\perp}$ and repeat the process.
Inductively, one obtains

$$V = U_1 \perp U_2 \perp \cdots \perp U_r \perp \operatorname{rad}(B),$$

where the U_i are one-dimensional nondegenerate subspaces.
Choosing a nonzero basis vector v_i from each U_i, together with a basis of the radical, produces a basis in which

$$G = \operatorname{diag}(d_1, \ldots, d_r, 0, \ldots, 0),$$

where

$$d_i \neq 0.$$

This is the basic congruence classification of symmetric bilinear forms before any assumptions on the field are used to normalize the nonzero diagonal entries.

5.17 Rank of a Bilinear Form

Definition 5.17.1. The rank of a bilinear form B is the rank of any representing matrix:

$$\operatorname{rank}(B) = \operatorname{rank}(G_{\mathcal{B}}(B)).$$

This is well defined because a change of basis replaces G by

$$P^T G P,$$

and multiplication by invertible matrices preserves rank.

If

$$\dim V = n,$$

then

$$\operatorname{rank}(B) = n - \dim \operatorname{rad}(B)$$

for a symmetric bilinear form.
Thus

$$B \text{ nondegenerate} \iff \operatorname{rank}(B) = n.$$

5.18 Classification over Different Fields

Diagonalization under congruence does not by itself imply that all symmetric bilinear forms have the same canonical form.
The field matters.
Over $\mathbb{C}$, every nonzero scalar has a square root. Therefore a nonzero diagonal coefficient d_i can be normalized to 1.
Hence every symmetric bilinear form over $\mathbb{C}$ is congruent to

$$\operatorname{diag}(\underbrace{1, \ldots, 1}_{r}, 0, \ldots, 0),$$

where $r = \operatorname{rank}(B)$.
Thus over $\mathbb{C}$, rank completely determines the congruence class of a symmetric bilinear form.
Over $\mathbb{R}$, positive and negative coefficients cannot be transformed into one another by multiplication by real squares. Therefore the signs of the diagonal coefficients carry additional information.
This leads to Sylvester's law of inertia, which will be developed in the next chapter together with quadratic forms.

5.19 Alternating Forms and Their Matrix Structure

Let B be alternating over a field with

$$\operatorname{char} K \neq 2.$$

Then its Gram matrix satisfies

$$G^T = -G.$$

A basic consequence is that the rank of an alternating form is even.

Theorem 5.19.1. The rank of a skew-symmetric matrix over a field of characteristic not equal to 2 is even.

Correspondingly, an alternating bilinear form can be decomposed into two-dimensional nondegenerate pieces together with its radical.
Its matrix may be reduced by congruence to block form

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus \cdots \oplus \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus 0.$$

This is the appropriate finite-dimensional linear algebra statement for alternating forms. We do not develop the geometric theories built from these forms in this volume.

5.20 Bilinear Forms and Inner Products

An inner product is a special kind of form, but the distinction is important.
Over $\mathbb{R}$, an inner product is a symmetric bilinear form satisfying positive definiteness:

$$B(x,x) > 0$$

for every $x \neq 0$.
A general symmetric bilinear form may satisfy

$$B(x,x) > 0$$

for some vectors,

$$B(x,x) < 0$$

for others, and may even have nonzero vectors satisfying

$$B(x,x) = 0.$$

Thus ordinary orthogonality theory is a special case of the more general theory of symmetric bilinear forms.
Over $\mathbb{C}$, an inner product is not bilinear but sesquilinear, because complex conjugation is required in one argument. That theory was already used in the inner product chapters and should not be confused with complex bilinear forms.

5.21 From Bilinear Forms to Quadratic Forms

For a symmetric bilinear form B, define

$$q(x) = B(x,x).$$

Then q is called the quadratic form associated with B.
If

$$B(x,y) = x^T A y$$

with $A = A^T$, then

$$q(x) = x^T A x.$$

Thus the matrix of a symmetric bilinear form also represents its associated quadratic form.
When

$$\operatorname{char} K \neq 2,$$

the bilinear form can be recovered from the quadratic form through the polarization identity

$$\boxed{B(x,y) = \frac{1}{2}\Big(q(x+y) - q(x) - q(y)\Big).}$$

Therefore symmetric bilinear forms and quadratic forms contain the same information over fields of characteristic not equal to 2.
The next chapter develops this correspondence systematically and focuses on the classification of real quadratic forms under congruence, including positive definiteness, inertia, and Sylvester's law of inertia.

Chapter 6

Quadratic Forms

6.1 Quadratic Forms

Let V be a finite-dimensional vector space over a field K, with

$$\operatorname{char} K \neq 2.$$

Definition 6.1.1. A quadratic form on V is a function

$$q : V \to K$$

for which there exists a symmetric bilinear form

$$B : V \times V \to K$$

such that

$$q(x) = B(x, x)$$

for every $x \in V$.

If

$$V = K^n$$

and

$$A = A^T \in M_n(K),$$

then

$$\boxed{q(x) = x^T A x}$$

defines a quadratic form.
Writing

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we obtain

$$q(x) = \sum_{i=1}^{n} a_{ii} x_i^2 + 2 \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j.$$

Thus the diagonal entries of A give the square terms, while the off-diagonal entries determine the mixed terms.

Example 6.1.2. The quadratic form

$$q(x_1, x_2) = 3x_1^2 + 4x_1 x_2 + 5x_2^2$$

has symmetric matrix

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix},$$

since

$$q(x) = x^T A x.$$

6.2 From Quadratic Forms to Symmetric Bilinear Forms

The associated symmetric bilinear form can be recovered from the quadratic form.

Theorem 6.2.1 (Polarization Identity)**.** Let

$$q(x) = B(x, x)$$

for a symmetric bilinear form B. Then

$$\boxed{B(x, y) = \frac{1}{2}\left(q(x + y) - q(x) - q(y)\right).}$$

Proof. Since B is symmetric,

$$q(x + y) = B(x + y, x + y).$$

Expanding,

$$q(x + y) = B(x, x) + 2B(x, y) + B(y, y).$$

Therefore

$$q(x + y) - q(x) - q(y) = 2B(x, y),$$

which gives the result. □

Equivalently,

$$B(x, y) = \frac{1}{4}\left(q(x + y) - q(x - y)\right).$$

Thus, when

$$\operatorname{char} K \neq 2,$$

quadratic forms and symmetric bilinear forms contain exactly the same information.

$$\boxed{\text{symmetric bilinear form} \longleftrightarrow \text{quadratic form.}}$$

6.3 Matrix Representation

Let

$$\mathcal{B} = (v_1, \ldots, v_n)$$

be a basis of V.
Suppose the symmetric bilinear form associated with q has Gram matrix

$$A = (a_{ij}).$$

Then for

$$x = \sum_{i=1}^{n} x_i v_i,$$

we have

$$q(x) = [x]_{\mathcal{B}}^T A [x]_{\mathcal{B}}.$$

Thus every quadratic form has a symmetric matrix representation.
Conversely, every symmetric matrix defines a quadratic form.

Remark 6.3.1. A nonsymmetric matrix may also appear in an expression

$$x^T A x,$$

but only its symmetric part contributes. Indeed,

$$x^T A x = x^T \left(\frac{A + A^T}{2}\right) x.$$

Hence quadratic forms may always be represented by symmetric matrices.

6.4 Change of Variables

Suppose

$$q(x) = x^T Ax$$

and introduce a new coordinate vector y by

$$x = Py,$$

where P is invertible.
Then

$$q(x) = (Py)^T A(Py).$$

Hence

$$q(x) = y^T (P^T AP)y.$$

Therefore the matrix of the quadratic form changes according to

$$\boxed{A \longmapsto P^T AP.}$$

This is precisely congruence.
Thus classifying quadratic forms under invertible linear changes of variables is equivalent to classifying symmetric matrices under congruence.

6.5 Diagonalization by Congruence

From the theory of symmetric bilinear forms, every symmetric matrix over a field of characteristic not equal to 2 is congruent to a diagonal matrix.
Therefore:

Theorem 6.5.1. For every quadratic form

$$q(x) = x^T Ax$$

with $A = A^T$, there exists an invertible matrix P such that

$$P^T AP = \operatorname{diag}(d_1, \ldots, d_n).$$

After the change of variables

$$x = Py,$$

the quadratic form becomes

$$\boxed{q = d_1 y_1^2 + \cdots + d_n y_n^2.}$$

Thus every quadratic form can be reduced to a sum of squares with scalar coefficients.
This is called a diagonal form of q.

6.6 Completing the Square

For explicit calculations, diagonalization can often be performed by completing squares.
Consider

$$q(x, y) = ax^2 + 2bxy + cy^2,$$

with $a \neq 0$.
Then

$$q(x, y) = a\left(x + \frac{b}{a}y\right)^2 + \left(c - \frac{b^2}{a}\right)y^2.$$

Thus the substitution

$$u = x + \frac{b}{a}y, \qquad v = y$$

eliminates the mixed term.

Example 6.6.1. Consider

$$q(x, y) = 2x^2 + 4xy + 5y^2.$$

Then

$$q(x, y) = 2(x + y)^2 + 3y^2.$$

With

$$u = x + y, \qquad v = y,$$

we obtain

$$q = 2u^2 + 3v^2.$$

In higher dimensions, the same idea can be repeated successively.

6.7 Congruence by Elementary Operations

Quadratic forms may also be reduced by matrix operations.
If

$$A \longmapsto P^T AP,$$

then every elementary change of variables corresponds to performing a column operation and the matching row operation.
For example:

- exchanging variables x_i and x_j corresponds to exchanging columns i, j and rows i, j;
- scaling a variable corresponds to scaling the corresponding row and column;
- replacing one variable by itself plus a multiple of another corresponds to compatible row and column operations.

This provides an efficient matrix method for diagonalizing quadratic forms under congruence.

6.8 Rank of a Quadratic Form

Definition 6.8.1. The rank of a quadratic form q is the rank of its associated symmetric matrix:

$$\operatorname{rank}(q) = \operatorname{rank}(A).$$

This is independent of the chosen basis because congruence preserves matrix rank.
If

$$q = d_1 y_1^2 + \cdots + d_r y_r^2$$

with

$$d_i \neq 0,$$

then

$$\operatorname{rank}(q) = r.$$

Thus rank is the number of nonzero terms in any diagonal form.

6.9 Real Quadratic Forms

From this point onward, let

$$K = \mathbb{R}.$$

Suppose a real quadratic form has been diagonalized:

$$q = d_1 y_1^2 + \cdots + d_n y_n^2.$$

Every nonzero coefficient can be normalized to either 1 or -1.
If

$$d_i > 0,$$

set

$$z_i = \sqrt{d_i}\, y_i.$$

If

$$d_i < 0,$$

set

$$z_i = \sqrt{-d_i}\, y_i.$$

Hence every real quadratic form is congruent to

$$\boxed{z_1^2 + \cdots + z_p^2 - z_{p+1}^2 - \cdots - z_{p+q}^2,}$$

possibly together with zero terms.
Equivalently, its matrix is congruent to

$$\operatorname{diag}(\underbrace{1, \ldots, 1}_{p}, \underbrace{-1, \ldots, -1}_{q}, \underbrace{0, \ldots, 0}_{r}).$$

Here

$$p + q + r = n.$$

6.10 Positive and Negative Indices

Definition 6.10.1. For a real quadratic form in canonical form

$$q = y_1^2 + \cdots + y_p^2 - y_{p+1}^2 - \cdots - y_{p+q}^2,$$

the integer p is called the positive index of inertia, and q is called the negative index of inertia.

The number of zero terms is

$$r = n - p - q.$$

The triple

$$\boxed{(p, q, r)}$$

is called the inertia of the quadratic form.
Its rank is

$$\operatorname{rank}(q) = p + q.$$

For a nondegenerate quadratic form,

$$r = 0.$$

6.11 Sylvester's Law of Inertia

The remarkable fact is that the integers p, q, r do not depend on the particular sequence of coordinate changes used to diagonalize the form.

Theorem 6.11.1 (Sylvester's Law of Inertia)**.** Let $A \in M_n(\mathbb{R})$ be symmetric. Suppose

$$A$$

is congruent to

$$\operatorname{diag}(I_p, -I_q, 0_r).$$

Then the integers

$$p, \qquad q, \qquad r$$

are uniquely determined by A.
In particular, they are invariant under real congruence.

Thus if two real symmetric matrices A and B are congruent, they have the same inertia. Conversely, two real symmetric matrices having the same inertia are congruent.
Therefore:

$$\boxed{A \text{ and } B \text{ are congruent over } \mathbb{R} \iff \operatorname{Inertia}(A) = \operatorname{Inertia}(B).}$$

This is the complete congruence classification of real symmetric matrices.
Your original notes record precisely this point: after choosing a basis, the Gram matrix can be reduced to 1's, −1's, and 0's, and the numbers of these entries are independent of the chosen basis. :contentReference[oaicite:2]index=2

6.12 Signature

There are two common conventions for the term signature.
In these notes, define the signature of a real quadratic form by

$$\boxed{\operatorname{sgn}(q) = p - q.}$$

Some texts instead use the ordered pair

$$(p, q)$$

and call this pair the signature.
To avoid ambiguity, the full invariant will normally be recorded as the inertia

$$(p, q, r).$$

For a nondegenerate form,

$$r = 0,$$

so

$$p + q = n.$$

In that case, rank and signature determine p and q:

$$p = \frac{n + \operatorname{sgn}(q)}{2},$$

$$q = \frac{n - \operatorname{sgn}(q)}{2}.$$

6.13 Geometric Meaning of Inertia

The inertia has a geometric interpretation.
Let

$$q(x) = x^T Ax$$

be a real quadratic form.
The positive index p is the largest possible dimension of a subspace $U \subseteq \mathbb{R}^n$ on which

$$q(x) > 0$$

for every nonzero $x \in U$.
Similarly, the negative index q is the largest possible dimension of a subspace W on which

$$q(x) < 0$$

for every nonzero $x \in W$.
The nullity r is

$$\dim \ker A.$$

Thus the inertia records how many independent positive, negative, and degenerate directions the quadratic form possesses.

6.14 Definite and Indefinite Quadratic Forms

Definition 6.14.1. A real quadratic form q is positive definite if

$$q(x) > 0$$

for every $x \neq 0$.

It is positive semidefinite if

$$q(x) \geq 0$$

for every x.
Similarly, q is negative definite if

$$q(x) < 0$$

for every $x \neq 0$, and negative semidefinite if

$$q(x) \leq 0$$

for every x.
A quadratic form is indefinite if it takes both positive and negative values.
In terms of inertia:

$$q \text{ positive definite} \iff (p, q, r) = (n, 0, 0),$$

$$q \text{ positive semidefinite} \iff q = 0 \text{ in the negative index,}$$

more explicitly,

$$q = 0, \qquad p + r = n.$$

Similarly,

$$q \text{ negative definite} \iff (p, q, r) = (0, n, 0).$$

A form is indefinite if

$$p > 0 \qquad \text{and} \qquad q > 0.$$

6.15 Eigenvalue Criterion for Definiteness

Let

$$A = A^T \in M_n(\mathbb{R}).$$

By the real spectral theorem,

$$A = QDQ^T,$$

where Q is orthogonal and

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Then

$$x^T Ax = y^T Dy, \qquad y = Q^T x.$$

Hence

$$q(x) = \sum_{i=1}^{n} \lambda_i y_i^2.$$

Therefore:

Theorem 6.15.1. Let A be real symmetric.

1. A is positive definite if and only if every eigenvalue of A is positive.
2. A is positive semidefinite if and only if every eigenvalue is nonnegative.
3. A is negative definite if and only if every eigenvalue is negative.
4. A is indefinite if and only if A has both positive and negative eigenvalues.

Thus for a real symmetric matrix, its inertia can also be read from the signs of its eigenvalues.

6.16 Leading Principal Minors

For

$$A = (a_{ij}) \in M_n(\mathbb{R}),$$

let A_k denote the upper-left $k \times k$ principal submatrix:

$$A_k = \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{kk} \end{pmatrix}.$$

Define the leading principal minors

$$\Delta_k = \det A_k, \qquad k = 1, \ldots, n.$$

These determinants provide another criterion for positive definiteness.

6.17 Sylvester's Criterion

Theorem 6.17.1 (Sylvester's Criterion)**.** Let

$$A = A^T \in M_n(\mathbb{R}).$$

Then A is positive definite if and only if

$$\boxed{\Delta_1 > 0, \quad \Delta_2 > 0, \quad \ldots, \quad \Delta_n > 0.}$$

Thus positive definiteness can be checked without explicitly computing the eigenvalues.

Example 6.17.2. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Then

$$\Delta_1 = 2 > 0$$

and

$$\Delta_2 = \det A = 5 > 0.$$

Hence A is positive definite.

For negative definiteness, the corresponding criterion is

$$(-1)^k \Delta_k > 0$$

for every k.
Equivalently,

$$\Delta_1 < 0, \qquad \Delta_2 > 0, \qquad \Delta_3 < 0, \ldots.$$

6.18 Positive Semidefiniteness

Positive semidefiniteness requires more care.
The conditions

$$\Delta_k \geq 0$$

for all leading principal minors are not sufficient in general.
Instead:

Theorem 6.18.1. A real symmetric matrix A is positive semidefinite if and only if every principal minor of A is nonnegative.

Equivalently, by the spectral theorem,

$$A \geq 0 \iff \lambda_i \geq 0$$

for all eigenvalues.
Thus positive semidefiniteness can be characterized either spectrally or through principal minors.

6.19 Cholesky Factorization and Positive Definiteness

Positive definite matrices also admit a useful factorization.

Theorem 6.19.1 (Cholesky Factorization)**.** If

$$A = A^T > 0,$$

then there exists an invertible upper triangular matrix R, with positive diagonal entries, such that

$$\boxed{A = R^T R.}$$

Equivalently,

$$A = LL^T$$

for a lower triangular matrix L.
Then

$$x^T Ax = x^T R^T Rx = \|Rx\|^2 .$$

Since R is invertible,

$$x \neq 0 \Longrightarrow Rx \neq 0,$$

and therefore

$$x^T Ax > 0.$$

Thus Cholesky factorization gives a direct algebraic representation of positive definiteness. This decomposition will be revisited later in the summary of matrix factorizations.

6.20 Quadratic Forms and Optimization

Quadratic forms appear naturally in multivariable optimization.
Consider

$$f(x) = \frac{1}{2}x^T Ax - b^T x + c,$$

where

$$A = A^T.$$

Its gradient is

$$\nabla f(x) = Ax - b.$$

A critical point therefore satisfies

$$Ax = b.$$

If A is positive definite, this critical point is the unique global minimum.
Indeed, if

$$x_* = A^{-1}b,$$

then

$$f(x) = f(x_*) + \frac{1}{2}(x - x_*)^T A(x - x_*).$$

Since $A > 0$,

$$f(x) \geq f(x_*),$$

with equality only when

$$x = x_*.$$

Thus positive definite quadratic forms are precisely the local models of strict convexity in the quadratic setting.

6.21 Quadratic Forms and Second Derivatives

Let

$$f : \mathbb{R}^n \to \mathbb{R}$$

be twice differentiable, and suppose x_0 is a critical point.
The second-order Taylor approximation contains the quadratic form

$$\frac{1}{2}h^T H_f(x_0)h,$$

where $H_f(x_0)$ is the Hessian matrix.
Since the Hessian is symmetric, its definiteness determines the local behavior:

$$H_f(x_0) > 0 \Longrightarrow x_0 \text{ is a strict local minimum,}$$

$$H_f(x_0) < 0 \Longrightarrow x_0 \text{ is a strict local maximum,}$$

while an indefinite Hessian indicates a saddle point.
Thus the classification of quadratic forms also underlies the second-derivative test in multivariable calculus.

6.22 Quadratic Hypersurfaces

A quadratic form may also be viewed geometrically through equations such as

$$q(x) = 1$$

or more generally

$$q(x) + \ell(x) + c = 0,$$

where ℓ is linear.
After a suitable change of coordinates, the quadratic part can be reduced to canonical form.
For example,

$$x_1^2 + x_2^2 - x_3^2 = 1$$

and

$$x_1^2 + x_2^2 + x_3^2 = 1$$

have fundamentally different geometries because their quadratic forms have different inertia.
Thus inertia provides algebraic information underlying the classification of real quadratic curves and surfaces.
The companion textbook also connects homogeneous quadratic polynomials with quadratic hypersurfaces in affine space. :contentReference[oaicite:3]index=3

6.23 Congruence versus Orthogonal Diagonalization

A real symmetric matrix admits two related diagonalizations.
By congruence,

$$P^T A P = \operatorname{diag}(I_p, -I_q, 0).$$

By the spectral theorem,

$$Q^T A Q = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

where Q is orthogonal.
These results serve different purposes.
Orthogonal diagonalization preserves the actual eigenvalues.
Congruence does not preserve eigenvalues, but preserves only their sign pattern:

$$(p, q, r).$$

Thus

$$\boxed{\text{orthogonal similarity} \longrightarrow \text{eigenvalues},}$$

whereas

$$\boxed{\text{congruence} \longrightarrow \text{inertia.}}$$

This distinction is essential.
For example,

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

and

$$\begin{pmatrix} 3 & 0 \\ 0 & 7 \end{pmatrix}$$

have different eigenvalues but are congruent over $\mathbb{R}$, since both are positive definite.

6.24 Summary of Real Quadratic Forms

For a real symmetric matrix A, the quadratic form

$$q(x) = x^T A x$$

can be studied through several equivalent viewpoints.
Its congruence canonical form is

$$\operatorname{diag}(I_p, -I_q, 0_r).$$

Its inertia is

$$(p, q, r).$$

Its rank is

$$p + q.$$

Its signature is

$$p - q.$$

Its definiteness is determined by the signs of its eigenvalues.
Positive definiteness may also be characterized by Sylvester's criterion:

$$\Delta_1 > 0, \ldots, \Delta_n > 0.$$

Thus the main chain of this chapter is

$$\boxed{\text{quadratic form} \longrightarrow \text{symmetric matrix} \longrightarrow \text{congruence} \longrightarrow \text{diagonal form} \longrightarrow \text{inertia.}}$$

Sylvester's law of inertia states that the final numbers of positive, negative, and zero directions are intrinsic and do not depend on the chosen coordinates.
Having completed the theory of bilinear and quadratic forms, the remaining part of the volume returns to matrices and develops factorizations that are useful both structurally and computationally. In particular, the singular value decomposition will combine the spectral theory of positive operators with orthogonal changes of coordinates.

Chapter 7

Singular Value Decomposition

7.1 Why Singular Values?

Eigenvalue decomposition is one of the most useful tools for studying square matrices. However, it has several limitations.
A matrix may fail to be diagonalizable. A rectangular matrix does not even have an ordinary eigenvalue problem of the form

$$Ax = \lambda x.$$

The singular value decomposition avoids both difficulties.
For every matrix

$$A \in K^{m\times n}, \qquad K = \mathbb{R} \text{ or } \mathbb{C},$$

there exist orthonormal coordinate systems in the domain and codomain in which A acts simply by independent nonnegative scalings.
The resulting decomposition has the form

$$\boxed{A = U\Sigma V^*.}$$

Unlike diagonalization, the singular value decomposition exists for every matrix.

7.2 The Matrices A^*A and AA^*

Let

$$A \in K^{m\times n}.$$

Then

$$A^*A \in K^{n\times n}$$

and

$$AA^* \in K^{m\times m}.$$

Both matrices are self-adjoint:

$$(A^*A)^* = A^*A$$

and

$$(AA^*)^* = AA^*.$$

They are also positive semidefinite.
Indeed, for every $x \in K^n$,

$$x^*A^*Ax = (Ax)^*(Ax) = \|Ax\|^2 \geq 0.$$

Thus

$$A^*A \geq 0.$$

Similarly,

$$AA^* \geq 0.$$

By the spectral theorem, both matrices are unitarily diagonalizable and have nonnegative eigenvalues.

7.3 Singular Values

Definition 7.3.1. Let

$$A \in K^{m \times n}.$$

The singular values of A are the nonnegative square roots of the eigenvalues of

$$A^*A.$$

Thus if the eigenvalues of A^*A are

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0,$$

then the singular values are

$$\sigma_i = \sqrt{\lambda_i}.$$

We usually arrange them in decreasing order:

$$\boxed{\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,}$$

followed by zero singular values if necessary.

Remark 7.3.2. Singular values are always real and nonnegative, even when A is a complex matrix or has complex eigenvalues.

7.4 Right Singular Vectors

Since A^*A is self-adjoint, there exists an orthonormal basis

$$v_1, \ldots, v_n$$

of K^n consisting of eigenvectors of A^*A.
Suppose

$$A^*Av_i = \sigma_i^2 v_i.$$

The vectors v_i are called right singular vectors of A.
For every i,

$$\|Av_i\|^2 = v_i^* A^* A v_i.$$

Hence

$$\|Av_i\|^2 = \sigma_i^2 \|v_i\|^2.$$

Since

$$\|v_i\| = 1,$$

we obtain

$$\boxed{\|Av_i\| = \sigma_i.}$$

Thus the singular value σ_i measures the amount by which A stretches the direction v_i.

7.5 Left Singular Vectors

Suppose

$$\sigma_i > 0.$$

Define

$$u_i = \frac{Av_i}{\sigma_i}.$$

Then

$$\|u_i\| = 1.$$

Moreover,

$$Av_i = \sigma_i u_i.$$

Applying A^*,

$$A^*u_i = \frac{A^*Av_i}{\sigma_i} = \sigma_i v_i.$$

Hence

$$\boxed{Av_i = \sigma_i u_i, \qquad A^*u_i = \sigma_i v_i.}$$

The vectors u_i are called left singular vectors.

Proposition 7.5.1. Left singular vectors corresponding to positive singular values can be chosen orthonormal.

Proof. For $i \neq j$,

$$\langle u_i, u_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle Av_i, Av_j \rangle .$$

Hence

$$\langle u_i, u_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle v_i, A^*Av_j \rangle .$$

Since

$$A^*Av_j = \sigma_j^2 v_j,$$

we obtain

$$\langle u_i, u_j \rangle = \frac{\sigma_j}{\sigma_i} \langle v_i, v_j \rangle = 0.$$

□

7.6 The Singular Value Decomposition

We now obtain the main theorem.

Theorem 7.6.1 (Singular Value Decomposition)**.** Let

$$A \in K^{m\times n}, \qquad K = \mathbb{R} \text{ or } \mathbb{C}.$$

Then there exist unitary matrices

$$U \in K^{m\times m}, \qquad V \in K^{n\times n},$$

and an $m \times n$ matrix Σ of the form

$$\Sigma = \begin{pmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & \\ & & \ddots & \\ 0 & & & \sigma_r \\ & & 0 & \end{pmatrix},$$

where

$$\sigma_1 \geq \cdots \geq \sigma_r > 0,$$

such that

$$\boxed{A = U\Sigma V^*.}$$

Over $\mathbb{R}$, the matrices U and V may be chosen orthogonal, so

$$A = U\Sigma V^T.$$

Proof. Choose an orthonormal eigenbasis

$$v_1, \ldots, v_n$$

of A^*A, ordered so that

$$A^*Av_i = \sigma_i^2 v_i$$

with

$$\sigma_1 \geq \cdots \geq \sigma_r > 0$$

and

$$\sigma_{r+1} = \cdots = \sigma_n = 0.$$

For $i \leq r$, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

Then

$$u_1, \ldots, u_r$$

are orthonormal.
Extend them to an orthonormal basis

$$u_1, \ldots, u_m$$

of K^m.
Set

$$U = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_m \\ | & & | \end{pmatrix}, \qquad V = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}.$$

Then

$$Av_i = \sigma_i u_i$$

for $i \leq r$, while

$$Av_i = 0$$

for $i > r$.
Therefore

$$AV = U\Sigma,$$

and hence

$$A = U\Sigma V^*.$$

□

The SVD listed in the original notes has exactly this form:

$$A = U\Sigma V^*.$$

:contentReference[oaicite:1]index=1

7.7 Compact Singular Value Decomposition

If

$$\operatorname{rank}(A) = r,$$

only the positive singular values contribute to A.
Define

$$U_r = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_r \\ | & & | \end{pmatrix},$$

$$V_r = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_r \\ | & & | \end{pmatrix},$$

and

$$\Sigma_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r).$$

Then

$$\boxed{A = U_r \Sigma_r V_r^*.}$$

This is called the compact or reduced SVD.
Here

$$U_r^* U_r = I_r, \qquad V_r^* V_r = I_r.$$

The compact SVD contains all nonzero information of the full SVD.

7.8 Rank and Singular Values

Theorem 7.8.1. The rank of A equals the number of positive singular values.

Thus

$$\boxed{\operatorname{rank}(A) = \#\{i : \sigma_i > 0\}.}$$

Indeed,

$$\operatorname{rank}(A) = \operatorname{rank}(A^*A).$$

Since A^*A is diagonalizable, its rank equals the number of nonzero eigenvalues, which is exactly the number of positive singular values.
Therefore

$$A \text{ is invertible } \iff \sigma_n > 0$$

for a square $n \times n$ matrix.

7.9 The Four Fundamental Subspaces

The SVD gives orthonormal bases adapted to the kernel and image of A.
Suppose

$$\operatorname{rank}(A) = r.$$

Then

$$Av_i = \sigma_i u_i$$

for $i \leq r$, while

$$Av_i = 0$$

for $i > r$.
Therefore

$$\boxed{\ker A = \operatorname{span}\{v_{r+1}, \ldots, v_n\}.}$$

Also,

$$\boxed{\operatorname{Im} A = \operatorname{span}\{u_1, \ldots, u_r\}.}$$

Since

$$A^* u_i = \sigma_i v_i,$$

we similarly obtain

$$\boxed{\operatorname{Im} A^* = \operatorname{span}\{v_1, \ldots, v_r\},}$$

and

$$\boxed{\ker A^* = \operatorname{span}\{u_{r+1}, \ldots, u_m\}.}$$

Hence the SVD gives the orthogonal decompositions

$$K^n = \operatorname{Im} A^* \oplus \ker A,$$

and

$$K^m = \operatorname{Im} A \oplus \ker A^*.$$

7.10 Spectral Decomposition of A^*A

Let

$$A = U\Sigma V^*.$$

Then

$$A^*A = V\Sigma^*\Sigma V^*.$$

Therefore

$$A^*A = V \operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)V^*.$$

Similarly,

$$AA^* = U\Sigma\Sigma^* U^*,$$

so

$$AA^* = U \operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)U^*.$$

Thus A^*A and AA^* have the same nonzero eigenvalues, including multiplicities.

$$\boxed{\operatorname{Spec}(A^*A) \setminus \{0\} = \operatorname{Spec}(AA^*) \setminus \{0\}.}$$

7.11 Geometric Interpretation

The SVD admits a particularly clear geometric interpretation.
For

$$x \in K^n,$$

write

$$x = Vy.$$

Then

$$Ax = U\Sigma y.$$

Thus the transformation A can be viewed as three successive operations:

$$\boxed{x \xrightarrow{V^*} y \xrightarrow{\Sigma} \Sigma y \xrightarrow{U} Ax.}$$

The first step is an orthogonal or unitary change of coordinates.
The second performs independent scalings:

$$y_i \longmapsto \sigma_i y_i.$$

The third is another orthogonal or unitary transformation.
Hence

$$\boxed{A = \text{rotation} \circ \text{scaling} \circ \text{rotation}.}$$

More precisely, in the real case the orthogonal transformations may include reflections.

7.12 Images of the Unit Sphere

Consider the unit sphere

$$\|x\| = 1.$$

Under the map

$$x \mapsto Ax,$$

the unit sphere is sent to an ellipsoid.
The principal axes of the ellipsoid are the left singular vectors

$$u_1, \ldots, u_r,$$

and the lengths of the corresponding semiaxes are

$$\sigma_1, \ldots, \sigma_r.$$

Thus singular values have a direct geometric meaning:

$$\boxed{\sigma_i = \text{principal stretching factor of } A.}$$

If some singular values vanish, the image collapses into a lower-dimensional subspace.

7.13 Largest and Smallest Singular Values

For every unit vector x,

$$\|Ax\|^2 = x^* A^* A x.$$

By the Rayleigh quotient characterization,

$$\max_{\|x\|=1} x^* A^* A x = \sigma_1^2.$$

Therefore

$$\boxed{\sigma_1 = \max_{\|x\|=1} \|Ax\|.}$$

Hence the largest singular value is the maximum stretching factor.
If A has full column rank, then

$$\boxed{\sigma_n = \min_{\|x\|=1} \|Ax\|.}$$

Thus the smallest singular value measures the minimum amount by which A stretches a unit vector.
For a square matrix,

$$A \text{ singular} \iff \sigma_{\min}(A) = 0.$$

7.14 Operator Norm

The matrix norm induced by the Euclidean norm is

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}.$$

Therefore

$$\boxed{\|A\|_2 = \sigma_1(A).}$$

Since

$$\|A^* A\|_2 = \sigma_1^2,$$

we also obtain

$$\|A\|_2^2 = \|A^* A\|_2.$$

For an invertible square matrix,

$$\left\|A^{-1}\right\|_2 = \frac{1}{\sigma_n(A)}.$$

Hence

$$\|A\|_2 \left\|A^{-1}\right\|_2 = \frac{\sigma_1(A)}{\sigma_n(A)}.$$

This ratio measures how differently the matrix stretches different directions.

7.15 Frobenius Norm and Singular Values

Recall the Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2}.$$

Since

$$\|A\|_F^2 = \operatorname{tr}(A^* A),$$

and the eigenvalues of A^*A are σ_i^2, we obtain

$$\boxed{\|A\|_F^2 = \sum_{i=1}^{r} \sigma_i^2.}$$

Thus

$$\|A\|_F = \left(\sum_{i=1}^{r} \sigma_i^2 \right)^{1/2}.$$

The Frobenius norm therefore measures the total squared stretching across all singular directions.

7.16 Rank-One Expansion

From the compact SVD,

$$A = U_r \Sigma_r V_r^*.$$

Expanding the product gives

$$\boxed{A = \sum_{i=1}^{r} \sigma_i u_i v_i^*.}$$

Each matrix

$$u_i v_i^*$$

has rank one.
Thus every rank-r matrix is a sum of r orthogonal rank-one components weighted by singular values.
This form is particularly important in approximation theory.

7.17 Low-Rank Approximation

Suppose

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^*,$$

with

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

For $k < r$, define the truncated SVD

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*.$$

Then

$$\operatorname{rank}(A_k) \leq k.$$

The discarded part is

$$A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^*.$$

Theorem 7.17.1 (Eckart–Young)**.** Among all matrices B satisfying

$$\operatorname{rank}(B) \leq k,$$

the truncated SVD gives a best approximation to A.
In the operator norm,

$$\boxed{\min_{\operatorname{rank}(B)\leq k} \|A - B\|_2 = \sigma_{k+1}.}$$

In the Frobenius norm,

$$\boxed{\min_{\operatorname{rank}(B)\leq k} \|A - B\|_F = \left(\sum_{i=k+1}^{r} \sigma_i^2 \right)^{1/2}.}$$

Thus small singular values correspond to directions that may be discarded with relatively little error.

7.18 The Moore–Penrose Pseudoinverse

Let

$$A = U\Sigma V^*$$

be an SVD.
Define Σ^+ by replacing each nonzero singular value σ_i with $1/\sigma_i$ and transposing the rectangular shape.

Definition 7.18.1. The Moore–Penrose pseudoinverse of A is

$$\boxed{A^+ = V\Sigma^+ U^*.}$$

If A is square and invertible, then

$$A^+ = A^{-1}.$$

More generally,

$$AA^+$$

is the orthogonal projection onto

$$\operatorname{Im} A,$$

while

$$A^+A$$

is the orthogonal projection onto

$$\operatorname{Im} A^*.$$

Thus the pseudoinverse extends the notion of inverse to singular and rectangular matrices.

7.19 Least Squares via the SVD

Consider

$$Ax = b.$$

When no exact solution exists, we seek to minimize

$$\|Ax - b\|.$$

Using

$$A = U\Sigma V^*,$$

write

$$x = Vy$$

and

$$c = U^*b.$$

Then

$$\|Ax - b\| = \|\Sigma y - c\|.$$

The problem separates into scalar equations in singular coordinates.
The minimum-norm least-squares solution is

$$\boxed{x = A^+b.}$$

Thus SVD provides a unified solution to least-squares problems even when A is rank deficient.

7.20 Singular Values and Determinants

Suppose

$$A \in M_n(K).$$

Then

$$\det(A^*A) = |\det A|^2.$$

On the other hand,

$$\det(A^*A) = \prod_{i=1}^{n} \sigma_i^2.$$

Therefore

$$\boxed{|\det A| = \prod_{i=1}^{n} \sigma_i.}$$

Geometrically, this means that the volume scaling factor of A is the product of its principal stretching factors.
If any singular value vanishes, then

$$\det A = 0.$$

7.21 SVD versus Eigenvalue Decomposition

The eigenvalue decomposition and SVD should not be confused.
For a diagonalizable square matrix,

$$A = PDP^{-1},$$

the same coordinate system is used for both the domain and codomain.
By contrast,

$$A = U\Sigma V^*$$

uses one orthonormal basis

$$v_1, \ldots, v_n$$

for the domain and another

$$u_1, \dots, u_m$$

for the codomain.
Eigenvalues may be negative or complex.
Singular values satisfy

$$\sigma_i \geq 0.$$

Eigenvalue decomposition may fail to exist.
The SVD always exists.
Thus:

$$\boxed{\text{eigendecomposition} = \text{similarity structure},}$$

while

$$\boxed{\text{SVD} = \text{orthogonal/unitary left-right structure.}}$$

7.22 SVD of a Normal Matrix

Suppose

$$A = UDU^*$$

is normal, where

$$D = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Then

$$A^*A = U \operatorname{diag}(|\lambda_1|^2, \dots, |\lambda_n|^2)U^*.$$

Therefore the singular values are

$$\boxed{\sigma_i = |\lambda_i|}$$

up to ordering.
For a positive semidefinite matrix,

$$\lambda_i \geq 0,$$

so its singular values are exactly its eigenvalues.
For a general Hermitian matrix, the singular values are the absolute values of its eigenvalues.

7.23 Polar Decomposition

The SVD immediately yields another important factorization.
Suppose

$$A = U\Sigma V^*.$$

Define

$$P = V\Sigma^*\Sigma^{1/2}V^*.$$

More directly,

$$\boxed{P = (A^*A)^{1/2}.}$$

In finite-dimensional notation, using the singular values,

$$P = V \operatorname{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)V^*.$$

Then P is positive semidefinite.

For a square invertible matrix, define

$$Q = UV^*.$$

Then Q is unitary and

$$\boxed{A = QP.}$$

This is the polar decomposition.
Similarly,

$$A = P'Q,$$

where

$$P' = (AA^*)^{1/2}.$$

Thus the matrix is decomposed into a unitary or orthogonal part and a positive part.
The original notes explicitly derive polar decomposition from the SVD and summarize it as

$$A = OP \quad \text{or} \quad A = PO.$$

:contentReference[oaicite:2]index=2 :contentReference[oaicite:3]index=3

7.24 Geometric Meaning of Polar Decomposition

For an invertible real matrix,

$$A = QP,$$

where Q is orthogonal and P is symmetric positive definite.
The positive factor P stretches space along mutually orthogonal directions.
The orthogonal factor Q then rotates or reflects the result.
Hence

$$\boxed{\text{general invertible transformation} = \text{pure stretch} + \text{rigid motion},}$$

where the composition is multiplicative rather than additive.
This is analogous to writing a nonzero complex number as

$$z = re^{i\theta},$$

with $r > 0$ representing size and $e^{i\theta}$ representing rotation.

7.25 SVD as a Culmination of Spectral Theory

The SVD is built almost entirely from ideas developed earlier in this volume.
Starting with a general matrix A, form

$$A^*A.$$

This matrix is self-adjoint and positive semidefinite. The spectral theorem gives an orthonormal eigenbasis:

$$A^*Av_i = \sigma_i^2 v_i.$$

From these vectors, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

The resulting two orthonormal systems produce

$$A = U\Sigma V^*.$$

Thus the logical chain is

$$\boxed{A \longrightarrow A^*A \longrightarrow \text{spectral theorem} \longrightarrow \text{singular values} \longrightarrow \text{SVD.}}$$

The decomposition simultaneously describes rank, kernel, image, operator norm, least squares, low-rank approximation, and geometric stretching.
For this reason, the singular value decomposition is one of the natural endpoints of finite-dimensional linear algebra.
The next chapter collects the major matrix decompositions appearing throughout the two volumes and compares their hypotheses, forms, and structural meanings.

Chapter 8

Matrix Decompositions

8.1 Why Matrix Decompositions?

A recurring principle throughout linear algebra is that a complicated matrix becomes easier to understand after it is replaced by simpler matrices.
Different problems lead to different notions of simplification.
For a linear operator, one studies similarity:

$$A \longmapsto P^{-1}AP.$$

This leads to diagonalization and Jordan normal form.
For an inner product space, one prefers orthogonal or unitary changes of basis:

$$A \longmapsto U^*AU.$$

This leads to Schur triangularization and the spectral theorem.
For a quadratic form, the natural transformation is congruence:

$$A \longmapsto P^TAP.$$

For a general rectangular matrix, different coordinate systems may be chosen in the domain and codomain:

$$A \longmapsto U^*AV.$$

This leads to the singular value decomposition.
Finally, many computational problems do not require a canonical form at all. Instead, one factors a matrix into simpler pieces:

$$A = LU, \qquad A = QR, \qquad A = R^*R,$$

and so on.
Thus the general principle is

$$\boxed{\text{matrix} \longrightarrow \text{structured factors} \longrightarrow \text{simpler computation or interpretation.}}$$

8.2 Several Different Types of Decomposition

It is useful to distinguish two broad ideas.
A *canonical form* replaces a matrix by a particularly simple matrix under an equivalence relation. Examples include

$$P^{-1}AP = J$$

for Jordan normal form and

$$U^*AU = D$$

for the spectral theorem.
A *factorization* writes the original matrix as a product:

$$A = A_1A_2 \cdots A_r.$$

Examples include

$$A = LU,$$
$$A = QR,$$

and

$$A = U\Sigma V^*.$$

Some constructions can be interpreted in both ways. For example,

$$A = UDU^*$$

is simultaneously a unitary diagonalization and a factorization.
Therefore the phrase “matrix decomposition” will be used broadly in this chapter.

8.3 LU Decomposition

Gaussian elimination transforms a matrix into an upper triangular matrix. The elimination process itself can be recorded by a lower triangular matrix.

Definition 8.3.1. An LU decomposition of a square matrix A is a factorization

$$\boxed{A = LU,}$$

where L is lower triangular and U is upper triangular.

Frequently, L is normalized so that all diagonal entries are equal to 1:

$$L = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ l_{21} & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ l_{n1} & \cdots & l_{n,n-1} & 1 \end{pmatrix}.$$

The matrix U is the upper triangular matrix obtained by Gaussian elimination.

8.4 LU Decomposition from Elimination

Suppose elimination transforms A into U:

$$E_k \cdots E_2 E_1 A = U,$$

where the E_i are elementary elimination matrices.
Then

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U.$$

Define

$$L = E_1^{-1} E_2^{-1} \cdots E_k^{-1}.$$

Then

$$A = LU.$$

The multipliers used during Gaussian elimination appear as entries of L.
Thus LU decomposition is essentially Gaussian elimination written as a matrix factorization.

8.5 Example of LU Decomposition

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

Eliminate the entry below the first pivot:

$$R_2 \mapsto R_2 - 2R_1.$$

Then

$$U = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

The elimination multiplier is 2, so

$$L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Indeed,

$$LU = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = A.$$

8.6 Pivoting and the $PA = LU$ Form

Not every matrix admits an LU decomposition without row exchanges.
When Gaussian elimination requires pivoting, the row exchanges can be recorded by a permutation matrix P.
The resulting factorization is

$$\boxed{PA = LU.}$$

Equivalently,

$$A = P^T LU,$$

since

$$P^{-1} = P^T$$

for a permutation matrix.
This form is the standard practical version of LU decomposition.

Remark 8.6.1. The precise sufficient conditions for an LU decomposition without pivoting may be formulated using nonzero leading principal minors. For computational purposes, the permutation form $PA = LU$ is more robust.

8.7 Solving Linear Systems with LU

Suppose

$$A = LU.$$

To solve

$$Ax = b,$$

write

$$LUx = b.$$

First solve

$$Ly = b$$

by forward substitution.
Then solve

$$Ux = y$$

by backward substitution.
Thus one difficult system is replaced by two triangular systems.
The advantage becomes especially important when several systems

$$Ax = b_1, \ldots, Ax = b_r$$

share the same coefficient matrix A. The LU decomposition is computed once and then reused for every right-hand side.

8.8 Rank Factorization

Let

$$A \in K^{m \times n}$$

have rank r.

Theorem 8.8.1 (Rank Factorization)**.** There exist matrices

$$C \in K^{m\times r}, \qquad F \in K^{r\times n},$$

such that

$$\boxed{A = CF,}$$

where

$$\operatorname{rank}(C) = \operatorname{rank}(F) = r.$$

One construction is obtained by choosing r linearly independent columns of A.
Let

$$C = \begin{pmatrix} | & & | \\ c_1 & \cdots & c_r \\ | & & | \end{pmatrix}$$

contain these basis columns.
Every column of A is a linear combination of $c_1, \ldots, c_r$. Place the corresponding coefficients into the columns of F. Then

$$A = CF.$$

Thus rank factorization separates the matrix into

$$K^n \xrightarrow{F} K^r \xrightarrow{C} K^m.$$

The intermediate dimension is exactly

$$r = \operatorname{rank}(A).$$

8.9 Meaning of Rank Factorization

Rank factorization makes the effective dimension of a linear map explicit.
If

$$A : K^n \to K^m$$

has rank r, then its action factors through an r-dimensional space:

$$\boxed{K^n \longrightarrow K^r \longrightarrow K^m.}$$

Thus the rank is not merely a number attached to a matrix. It is the minimum dimension through which the corresponding linear map can factor.
The column space satisfies

$$C(A) = C(C),$$

while the row space of A is represented by the rows of F.

8.10 QR Decomposition Revisited

The QR decomposition was obtained from the Gram–Schmidt process.
For a matrix

$$A \in K^{m\times n}$$

with linearly independent columns,

$$\boxed{A = QR,}$$

where the columns of Q are orthonormal and R is upper triangular.
Thus

$$Q^*Q = I.$$

If A is square and invertible, Q is unitary over $\mathbb{C}$ or orthogonal over $\mathbb{R}$.
The QR decomposition may therefore be interpreted as

$$\boxed{\text{arbitrary independent columns} \longrightarrow \text{orthonormal columns} + \text{triangular coefficients.}}$$

It is the matrix form of Gram–Schmidt orthogonalization.

8.11 QR and Least Squares

For

$$A = QR,$$

the least-squares problem

$$\min_x \|Ax - b\|$$

reduces to

$$Rx = Q^*b.$$

Hence QR decomposition converts a least-squares problem into an upper triangular system.
Compared with the normal equations

$$A^*Ax = A^*b,$$

the QR formulation works directly with the geometry of the column space.

8.12 Cholesky Decomposition Revisited

If

$$A = A^* > 0,$$

then A admits the Cholesky factorization

$$\boxed{A = R^*R,}$$

where R is upper triangular with positive diagonal entries.
Over $\mathbb{R}$,

$$A = R^TR.$$

Equivalently,

$$A = LL^*$$

for a lower triangular matrix L.
The factorization reflects positive definiteness:

$$x^*Ax = \|Rx\|^2.$$

Thus

$$A > 0$$

is represented as a Gram-type product.

8.13 Cholesky and Quadratic Forms

Suppose

$$A = R^T R$$

is real positive definite.
Then the associated quadratic form satisfies

$$q(x) = x^T Ax = \|Rx\|^2 .$$

Hence after the change of variables

$$y = Rx,$$

the quadratic form becomes

$$q(x) = y_1^2 + \cdots + y_n^2.$$

Thus Cholesky factorization is closely related to the congruence classification of positive definite quadratic forms.

8.14 Schur Decomposition Revisited

For every complex square matrix A, Schur triangularization gives

$$\boxed{A = URU^*,}$$

where U is unitary and R is upper triangular.
Equivalently,

$$U^*AU = R.$$

The diagonal entries of R are the eigenvalues of A.
Schur decomposition always exists over $\mathbb{C}$, regardless of whether A is diagonalizable.
Thus it lies conceptually between a general matrix and a diagonal spectral form:

$$\boxed{\text{general matrix} \longrightarrow \text{unitarily triangular} \longrightarrow \text{unitarily diagonal if normal.}}$$

8.15 Eigendecomposition

If A is diagonalizable, then

$$\boxed{A = PDP^{-1},}$$

where D is diagonal.
The columns of P are eigenvectors, and the diagonal entries of D are the corresponding eigenvalues.
This decomposition exists if and only if A has a basis of eigenvectors.
It is therefore more restrictive than Schur decomposition but simpler when available.
If A is normal, P may be chosen unitary:

$$A = UDU^*.$$

If A is real symmetric, P may be chosen orthogonal:

$$A = QDQ^T.$$

8.16 Jordan Decomposition Revisited

If the characteristic polynomial of A splits over K, then

$$\boxed{A = PJP^{-1},}$$

where J is in Jordan normal form.
The decomposition exists even when A is not diagonalizable.
Each Jordan block has the form

$$J_k(\lambda) = \lambda I_k + N_k,$$

where N_k is nilpotent.
Thus Jordan normal form separates an operator into eigenvalue data and nilpotent defects.
Diagonalization is the special case in which every Jordan block has size 1.

8.17 SVD Revisited

For every matrix

$$A \in K^{m \times n},$$

the singular value decomposition gives

$$\boxed{A = U\Sigma V^*.}$$

Unlike eigendecomposition and Jordan form, SVD applies to rectangular matrices.
The singular values satisfy

$$\sigma_i \geq 0,$$

and the number of positive singular values equals

$$\operatorname{rank}(A).$$

The SVD uses one orthonormal basis for the domain and another for the codomain.
Therefore it is fundamentally a left–right decomposition rather than a similarity decomposition.

8.18 Polar Decomposition Revisited

For a square matrix A, define

$$P = (A^*A)^{1/2}.$$

If A is invertible, then

$$\boxed{A = UP,}$$

where U is unitary and P is positive definite.
More generally, for arbitrary A, the unitary factor is replaced by the appropriate partial isometry.
For an invertible real matrix,

$$A = QP,$$

where Q is orthogonal and P is symmetric positive definite.
Thus polar decomposition separates

$$\boxed{\text{geometry-preserving motion}}$$

from

$$\boxed{\text{positive stretching.}}$$

The original notes explicitly derive the polar decomposition from SVD. :contentReference[oaicite:2]index=2

8.19 Relationships among Decompositions

Several decompositions developed in this volume are closely related.
Starting from the SVD

$$A = U\Sigma V^*,$$

we obtain

$$A^*A = V\Sigma^*\Sigma V^*.$$

Thus the SVD is obtained by applying the spectral theorem to A^*A.
From the same SVD,

$$A = (UV^*)(V\Sigma V^*)$$

for an invertible square matrix, producing the polar decomposition.
Similarly, Gram–Schmidt applied to the columns of A produces the QR decomposition.
Gaussian elimination produces LU.
Positive definiteness specializes triangular factorization to Cholesky.
Normality specializes Schur triangularization to unitary diagonalization.
Thus many decompositions arise by adding additional structure to a more general situation.

8.20 A Hierarchy of Structural Assumptions

The hypotheses may be arranged schematically as follows.
Every complex square matrix admits

$$A = URU^*.$$

If A is normal, then

$$R = D,$$

so

$$A = UDU^*.$$

If A is Hermitian, then the entries of D are real.
If A is positive semidefinite, then

$$D \geq 0.$$

If A is positive definite, then

$$D > 0$$

and a Cholesky factorization exists:

$$A = R^*R.$$

Thus

$$\boxed{\text{positive definite} \subset \text{Hermitian} \subset \text{normal} \subset \text{all complex matrices},}$$

where each additional assumption gives a stronger decomposition.

8.21 Which Decomposition Should Be Used?

Different decompositions answer different questions.
If the objective is to solve

$$Ax = b$$

efficiently for many right-hand sides, LU is natural.
If orthogonality or least squares is central, QR is preferable.
If A is positive definite, Cholesky exploits this additional structure.
If eigenvalues are required for a general complex matrix, Schur form provides a stable triangular representation.
If the operator is diagonalizable, eigendecomposition gives the simplest similarity form.
If diagonalization fails, Jordan form describes the complete similarity structure.
If A is rectangular or low-rank structure matters, SVD is the natural decomposition.
If one wants to separate stretching from rigid motion, polar decomposition is appropriate.
Thus no decomposition is universally best.
The correct decomposition depends on the structure of the matrix and the question being asked.

8.22 Canonical Forms versus Computational Factorizations

It is useful to separate two goals.
Jordan normal form gives a complete theoretical invariant for similarity, but it is highly sensitive to perturbations and is not usually the preferred numerical representation.
By contrast, QR and Schur decompositions are particularly useful in numerical computation because they rely on orthogonal or unitary transformations.
Similarly, the SVD is both theoretically canonical and numerically useful.
Thus the theoretically simplest representation is not always the most stable computational representation.
This distinction becomes increasingly important in numerical linear algebra.

8.23 Summary Table of Matrix Decompositions

The principal decompositions appearing in these notes may be summarized as follows:

Decomposition	Form	Main Structure
LU	$A = LU$	Gaussian elimination
QR	$A = QR$	orthogonality
Cholesky	$A = R^*R$	positive definiteness
Schur	$A = URU^*$	unitary triangularization
Eigen	$A = PDP^{-1}$	diagonalizable operator
Jordan	$A = PJP^{-1}$	similarity classification
SVD	$A = U\Sigma V^*$	singular directions
Polar	$A = UP$	motion and stretching

The original course notes similarly collect LU, QR, rank factorization, Cholesky, Schur, eigendecomposition, SVD, polar decomposition, and Jordan form as major matrix decompositions. :contentReference[oaicite:3]index=3

8.24 The Structural Thread of the Second Volume

The second volume began with a limitation of ordinary diagonalization.

When eigenvectors were insufficient, generalized eigenvectors led to Jordan normal form:

$$\text{diagonalization} \longrightarrow \text{Jordan form}.$$

Adding an inner product introduced geometry:

$$\text{inner products} \longrightarrow \text{orthogonality} \longrightarrow \text{Gram–Schmidt}.$$

Adjoints then led to normal operators and the spectral theorem:

$$\text{adjoint} \longrightarrow \text{normality} \longrightarrow \text{unitary diagonalization}.$$

Bilinear and quadratic forms introduced a different equivalence:

$$\text{bilinear form} \longrightarrow \text{congruence} \longrightarrow \text{inertia}.$$

Finally, the singular value decomposition treated arbitrary rectangular matrices:

$$\text{spectral theorem for } A^*A \longrightarrow \text{SVD} \longrightarrow \text{polar decomposition}.$$

Thus the overall development is

$$\boxed{\text{canonical forms} \longrightarrow \text{geometric structure} \longrightarrow \text{spectral structure} \longrightarrow \text{matrix factorizations}.}$$

8.25 Completion of Undergraduate Advanced Linear Algebra

Together, the first and second volumes form a coherent finite-dimensional theory.
The first volume developed

$$\boxed{\text{linear systems} \longrightarrow \text{matrices} \longrightarrow \text{vector spaces} \longrightarrow \text{linear maps}}$$

and then

$$\boxed{\text{determinants} \longrightarrow \text{eigenvalues} \longrightarrow \text{diagonalization}.}$$

The second volume continued with

$$\boxed{\text{Jordan theory} \longrightarrow \text{inner product geometry} \longrightarrow \text{spectral theory}}$$

and then

$$\boxed{\text{bilinear forms} \longrightarrow \text{quadratic forms} \longrightarrow \text{SVD and matrix decompositions}.}$$

The unifying principle throughout both volumes is that the same linear object may have many coordinate representations. The correct notion of equivalence determines which properties are intrinsic and which simplifications are possible.
For linear operators, this notion is similarity.
For quadratic forms, it is congruence.
For inner product geometry, orthogonal and unitary transformations are the natural coordinate changes.
For general maps between two inner product spaces, independent orthonormal bases in the domain and codomain lead to the singular value decomposition.
Thus the central question of advanced linear algebra is not merely how to calculate with a matrix, but rather

$$\boxed{\text{What structure does the matrix represent, and what is the simplest coordinate description compatible}}$$

This concludes the core undergraduate advanced linear algebra sequence.
More abstract extensions—including modules, tensor products, exact sequences, and related algebraic structures—are separated into the companion volume *Algebraic Frontiers*.

Acknowledgements

I would like to express my sincere gratitude to Professor Ziqing Xiang for teaching the honors course in Advanced Linear Algebra. The course introduced many structural viewpoints beyond elementary matrix calculation and provided the original context from which much of the material in this volume was reorganized.

I am also grateful to Leo, the teaching assistant for this course, for his careful work throughout the semester. He also served as the teaching assistant for my Abstract Algebra course, and in both courses he was consistently responsible in grading assignments and quizzes, answering questions, and providing supplementary explanations and materials when needed.

Although I did not fully understand every topic when I first took the course, revisiting and reorganizing the material into these notes has helped me distinguish the core ideas of undergraduate advanced linear algebra from the more abstract algebraic topics that belong to later study.